

Additional Results and Reproduction Details

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This companion accompanies *When a High Score Is an Illusion: Certifying Genuine versus Repackaged Forecasting Skill*, revision 7. The main paper contains every core theoretical statement and proof. This document preserves the expanded comparison protocols, complete model tables, additional bounds and provenance qualifications. Results from distinct fitting rules remain separate. The code release provides aggregate evidence and executable public or synthetic analyses at <https://github.com/MaxwellNi/forecast-audit/releases/tag/icdm2026-v7>.

1 Corrections and interpretation of the stored forecasts

The historical row labelled TimesFM-2.5 was produced by a forecasting network whose adapter did not load the intended checkpoint. The table now calls it an *untrained forecasting network*; its stored predictions and numerical audit results are unchanged. A separate rerun used the appropriate pretrained architecture, loading checks and contexts ending at each forecast origin. These later aggregates describe different forecasts. Historical availability of the pretrained weights has not been established.

For 29 of the 30 historical forecasters, excess-return training targets used inconsistent risk-free-rate units. Subtracting a common monthly shift preserves within-month ranks mathematically for frozen outcomes. It does not preserve the model that would result from correct retraining; finite-precision rounding can also change ties. The restricted case remains an audit of stored outputs.

The retail price-and-calendar example has standard-fold statistic 7.68, ordered-middle statistic -0.82 , and ordered-all statistic -1.21 . Croston SBA has -8.59 , 6.11 , and -1.04 , respectively. Seven models are retained in each of the first two configurations, but their identities differ. Middle-fold evaluation changes support as well as nuisance fitting, so this comparison alone does not identify the causal share of feedback bias. An additional matched-middle comparison fixes evaluation rows: the price model changes from 6.41 to -0.82 and Croston SBA from -19.43 to 6.11 when only fitting direction changes; family retentions change from five to seven. This establishes sensitivity to the nuisance-fitting rule on that sample, not a population calibration result. The ratings example in the main paper supplies the accuracy-versus-association illustration without relying on stability of the retail model's retention.

A previously reported solver discrepancy bound of 3.2×10^{-10} has a located aggregate verification record: its largest recorded discrepancy is below that bound. It concerns replication of frozen audit calculations, not correctness of forecast training or statistical calibration.

2 Complete-family audit procedure

The main paper presents the information-to-action table. The executable procedure retains the following steps for every fixed family:

1. Fix forecasts, outcomes, target, sampling units, fitting rule, analysis scale and level.
2. For every model, record outcome-free redundancy and evaluation support.
3. Select the applicable information row and construct its residual score and preselected bound or statistic.
4. Record the null and assumptions for theorem-supported inputs; label other inputs as nominal screens.
5. Set $p^* = 1$ for recorded abstention, otherwise $p^* = p$.
6. Apply BY to the complete family, including abstentions.
7. Return abstain, retain or not retained, with target, validity status and all raw values.

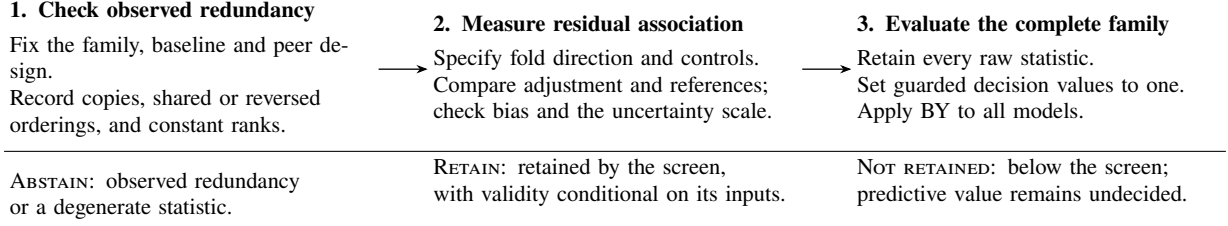


Figure 1: **From forecasts to an interpretable audit.** Keep the baseline, target and sampling design fixed. Report observed redundancy, residual association and complete-family decisions. FDR control requires valid input p -values.

3 Panel scale and redundancy details

3.1 Adjustment and the uncertainty scale

At resolution q , cross-fitted regressions on the represented controls leave residuals $e_X^{(q)}, e_Y^{(q)}$. The panel implementation uses additive entity effects and q baseline bins. Period enters ranking and clustering. This additive approximation need not equal the full joint conditional mean. Let $R_i^{(q)} = e_{X,i}^{(q)} e_{Y,i}^{(q)}$. The i.i.d. generalized covariance measure uses $T(q) = \sqrt{n} \bar{R}^{(q)} / \widehat{\text{sd}}(R^{(q)})$. For cluster inference and products s_i , let $L_g = \sum_{i \in g} s_i$, $n_g = |g|$, and

$$\widehat{\text{se}}_{\text{cl}}^2 = n^{-2} \sum_{g=1}^M (L_g - n_g \bar{s})^2, \quad T_{\text{cl}} = \bar{s} / \widehat{\text{se}}_{\text{cl}}. \quad (1)$$

Ordered screens use Newey–West variance with a stated Bartlett lag [1]; lag zero gives Eq. (1). Cluster GCM inference needs a cluster CLT, consistent variance and negligible fitted-product error [2, 3]; fixed lags and additive fits alone do not establish them.

3.2 Redundancy and resolution sensitivity

For unit-variance X , let $\rho^2 = R^2(X | Z)$. The Cauchy-Schwarz inequality gives

$$|\theta_\infty| \leq \sqrt{1 - \rho^2} \|Y - \mathbb{E}[Y | Z]\|_2. \quad (2)$$

High overlap limits covariance, not studentized power. Our outcome-free abstention rule records exact copies, matching or reversed cluster orders with ties, and constant ranks. Setting $p^* = 1$ on abstention preserves valid marginal inputs because $p^* \geq p$; it does not calibrate invalid ones.

Resolution sensitivity On a fixed grid $\mathcal{Q}$, let $s_i = \sum_q w_q R_i^{(q)}$ with $\sum_q w_q = 1$ and $\sum_q w_q q^{-\beta} = 0$. An expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ then leaves $O(q_{\min}^{-2\beta})$ bias for bounded weights. Inference requires this remainder and fitting error below the standard error. Smooth uniform controls give $\beta = 2$; Gaussian quantile bins instead have order $1/(q \log q)$. The companion proves both cases. We compare both fixed exponents with splines as screens, without choosing by significance.

Connection to classical rank correlation For continuous margins, $\Gamma = (3\tau - \rho_S)/12$, where τ and ρ_S are population Kendall and Spearman correlations. This is Hoeffding’s classical decomposition [4, 5]. The conditional audit separates that term from nuisance error and quantifies its amplification relative to sampling uncertainty.

For continuous margins, the population rank identity follows from $\mathbb{E}[a_{12}b_{12}] = (1 + \tau)/4$ and $\mathbb{E}[F_V(V)F_W(W)] = (3 + \rho_S)/12$.

4 Expanded figures and case protocols

A numerical audit For electricity DLinear with complementary all-fold fitting and exponent two, the mean product is 0.016425, its 14-lag HAC standard error is 0.001874 and $T = 8.763$ over 729 days. The nominal $p = 9.52 \times 10^{-19}$ enters the complete 11-model family; DLinear is retained. These are the recorded screen’s calculations, rather than a calibration check.

Table 1: Complete-family screens at 0.05, holding original copy flags fixed. All/middle denotes evaluation folds; complementary/directional denotes nuisance fitting. MovieLens orders user clusters, not time. Counts do not establish calibrated discoveries.

Task	Models	Complementary		Directional		Abstain
		All	Middle	All	Middle	
Electricity	11	10	10	10	10	1
M5	10	7	5	6	7	2
MovieLens-25M	10	6	6	6	6	4
OSAP forecasts	10	0	0	0	0	1

For ordered periods, let $A_t = L_t - n_t \bar{s}$ on the complete calendar of M_{cal} periods, with $A_t = 0$ for a missing period. At specified lag ℓ , the heteroskedasticity and autocorrelation consistent (HAC) estimator of Newey and West [1] is

$$\begin{aligned} \widehat{\text{se}}_{\text{HAC}}^2 = n^{-2} & \left[\sum_{t=1}^{M_{\text{cal}}} A_t^2 \right. \\ & \left. + 2 \sum_{h=1}^{\min(\ell, M_{\text{cal}}-1)} \left(1 - \frac{h}{\ell+1} \right) \sum_{t=h+1}^{M_{\text{cal}}} A_t A_{t-h} \right]. \end{aligned} \quad (3)$$

Lag zero recovers Eq. (1), without a degrees-of-freedom multiplier. Fixed lags define the reported screens.

Separate financial targets A two-sided factor-alpha audit using Fama and French factors and momentum [6, 7], with 12 HAC lags, retains 122 of 212 published signals. This tests another target and is not a fifth independent dataset. In the restricted forecast family, the single standard-fold nominal positive changes from $T = 2.45$ (BY-adjusted $p = 0.85$) to 1.29 with ordered middle folds; no model is retained. Among sixteen strongly negative standard statistics, median absolute magnitude falls from 3.34 to 1.08. These comparisons diagnose nuisance sensitivity rather than establish absence of predictive information.

Separate simulations on two observed incidence patterns use entity-and-period clustering. Standard all-fold fitting has largest absolute mean 3.092, versus 0.251 for directional middle-fold fitting on the larger pattern. This contrast also changes evaluation support. At lag-one outcome correlation 0.9, another stress test gives an ordered-versus-independent-fit mean contrast up to 6.47 standard units, outside the innovation assumption.

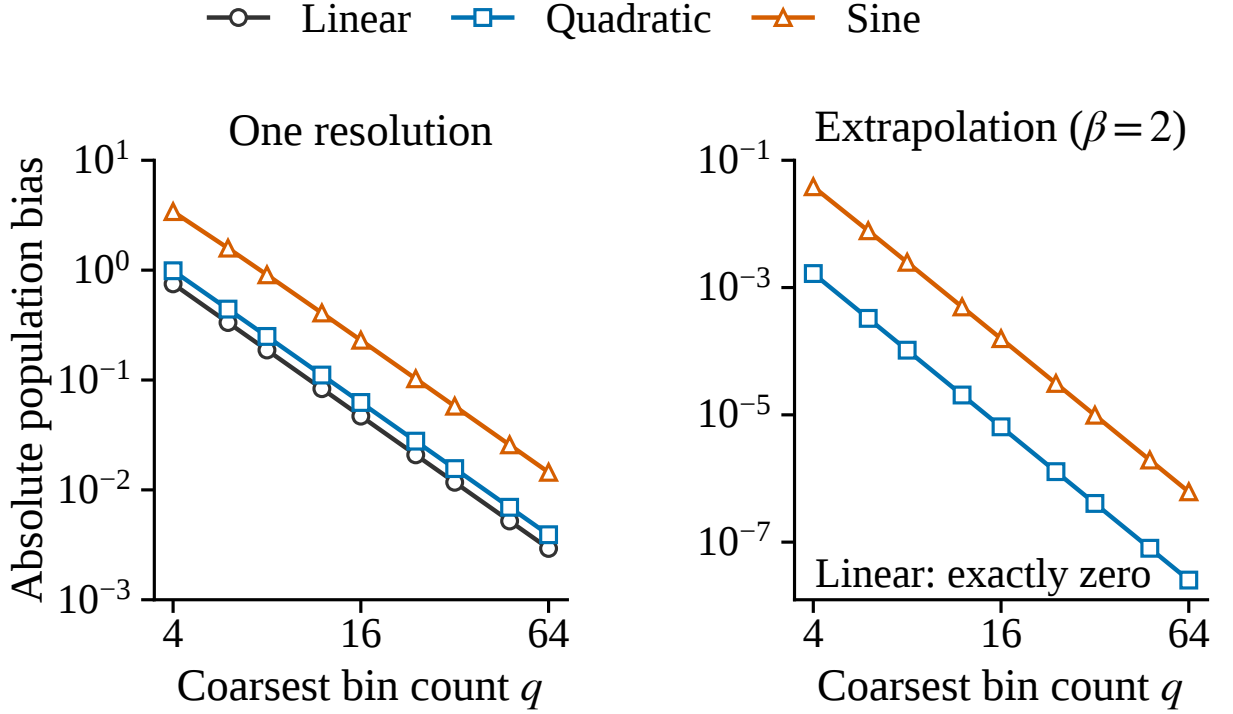


Figure 2: **Exact population approximation bias.** For $X = f(U) + \epsilon$, $Y = f(U) + \eta$ and uniform U , use $f(U) = 12U$, $12U^2$, or $6\sin(2\pi U)$. The right panel combines bins $q\{1, 2, 4\}$ with weights $(-1/6, 1/2, 2/3)$. The linear remainder is zero; the other remainders persist. These are oracle identities, not fitted simulation means.

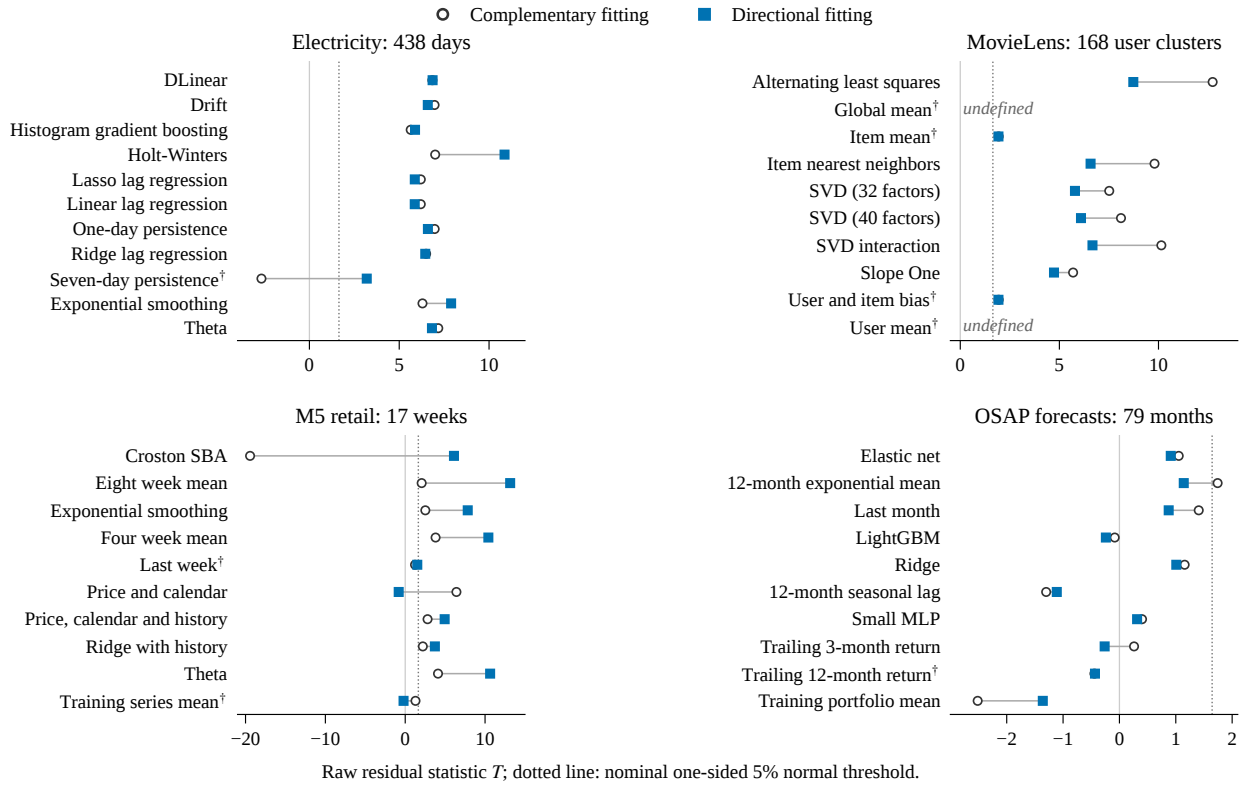


Figure 3: **Same forecasts and observations, different nuisance training.** All 41 models use the middle three folds, identical ranks and bins, $\beta = 2$ and the original uncertainty rule. Lines join raw statistics; the dotted line is the nominal 5% normal threshold. Daggers mark fixed full-cohort abstentions; two pairs are undefined. MovieLens orders user clusters. These are sensitivity comparisons, not calibrated discoveries.

Table 2: Scalar comparison, $n = 400$, 300 paired draws per cell. The first four rates test independent nulls; power is the minimum over four covariance alternatives. Zero covariance with dependence is an alternative for KCI, a null for the covariance tests.

Test	false-positive rate				min.	zero
	linear	smooth	quad.	t_5	power	cov.
GCM (8 bins)	0.220	0.070	0.987	0.090	0.980	0.060
GCM (32 bins)	0.140	0.097	0.590	0.120	0.957	0.097
GCM (splines)	0.100	0.070	0.113	0.093	0.947	0.050
Extrap. $\beta = 1$	0.120	0.093	0.210	0.107	0.943	0.093
Extrap. $\beta = 2$	0.140	0.090	0.613	0.107	0.967	0.077
KCI (gamma)	0.070	0.047	0.047	0.070	0.727	1.000

5 Expanded standard-fit comparison protocols

The following comparisons use the standard nuisance folds unless explicitly stated otherwise. They complement the distinct reference-sampling and feedback experiments in the main paper. Complete cell-level results, including unsuccessful settings, remain in the release.

5.1 Matched comparison on independent observations

All methods receive the same $n = 400$ observations per replication. Canonical KCI [8] uses `causal-learn` 0.1.4.8, Gaussian kernels, empirical bandwidths, regularization 0.001, and a gamma null approximation. Independent matrix calculations verify its statistic and gamma moments. GCM variants share two balanced folds and i.i.d. studentization. Adjustment uses eight or 32 bins, cubic splines with eight training quantile knots and ridge penalty 0.001, or the ladder $\{8, 12, 16, 24, 32\}$. The original comparison fixes $\beta = 1$; a separately recorded extension evaluates $\beta = 2$ on identical draws. Bins, knots and coefficients use training folds; KCI uses its native full sample.

Let Z, U, e_X, e_Y be independent standard normal variables. The common baseline is $g(Z) = 0.8Z, 0.8 \tanh(1.5Z)$, or $0.8(Z^2 - 1)$. The heavy-tail case uses the smooth baseline and independent variance-one t_5 errors. Under the null, $X = g(Z) + e_X$ and $Y = g(Z) + e_Y$. The positive-covariance alternative adds $0.5U$ to both, giving conditional covariance 0.25. A further alternative uses the smooth baseline and $Y = g(Z) + (e_X^2 - 1)/\sqrt{2} + e_Y$. Its conditional covariance with X is zero, but it is conditionally dependent.

Each of the nine cells uses 300 paired replications. Raw variables retain independent sampling. U-centered partial distance correlation [9], verified in the package, targets a different null.

5.2 Calibration and power

Table 2 and Fig. 4 show incomplete bias correction. At $\beta = 1$, extrapolation’s quadratic-null rate is $63/300 = 0.210$, with Wilson 95% interval $[0.168, 0.260]$. At $\beta = 2$, it rises to $184/300 = 0.613$, with interval $[0.557, 0.667]$. All four $\beta = 2$ null intervals lie above 0.05. Gaussian quantile bins need not satisfy the uniform-control expansion proved below; neither exponent is justified here by Proposition 1. KCI’s four rates range from 0.047 to 0.070; their individual intervals contain 0.05. These are pointwise intervals.

At $\beta = 1$, minimum power is 0.943, versus 0.727 for KCI. Unequal null rejection rates prevent ranking these raw powers at matched size. Splines improve both worst null rate and minimum power over $\beta = 1$ extrapolation. Under dependence with zero covariance, KCI rejects 300/300 and $\beta = 1$ extrapolation 28/300: the two tests address different nulls.

Comparison with supplied null simulations A retrospective secondary analysis assigns the first 150 null draws per regime to calibration and the remaining 150 to validation. Each method’s Monte Carlo value is $p_{MC} = (1 + \sum_{b=1}^{150} \mathbf{1}\{p_b \leq p_0\})/151$, where p_b and p_0 are the calibration and evaluation raw p -values. Exchangeability of the 151 null ranks, including ties, gives marginal rejection at most $7/151$ at level 0.05; rates conditional on a realized bank can differ. On the 300 independent positive draws, $\beta = 1$ detects 0.910/0.937 in the linear/smooth cases versus KCI’s 0.727/0.753; under quadratic controls, splines give 0.933 versus 0.853 for $\beta = 1$. The package retains all 48 cells and 60 paired contrasts, using exact two-sided binomial tests of discordant decisions and Holm adjustment. The supplied null law is extra information, not a real-panel calibration.

Table 3: Rough controls, 300 paired draws. K, S, O : kink, step, oscillation; 0: independent null; +: positive covariance. K_d has dependence with zero covariance, a null for GCM and alternative for KCI. Entries are rejection proportions.

Method	K_0	S_0	O_0	K_+	S_+	O_+	K_d
GCM, 8 bins	.180	.230	.810	.993	.993	1.000	.110
GCM, 32 bins	.120	.140	.217	.963	.983	.993	.097
Spline GCM	.070	.177	.157	.957	.983	.967	.080
Extrap., $\beta = 1$	.093	.120	.147	.947	.967	.980	.090
KCI	.057	.097	.037	.733	.863	.670	1.000

Table 4: Rejections among 300 paired panels with identical ranks, entities, folds and cluster scale. All 33 settings and their intervals are provided in the package.

Condition	M	$\beta = 1$	$\beta = 2$	Spline
Independent null	25	25	25	24
Independent null	50	18	17	17
Independent null	100	12	11	10
High overlap null	25	0	23	30
High overlap null	50	0	15	17
High overlap null	100	0	7	10
Positive control	25	300	300	300
Positive control	50	300	300	300
Positive control	100	300	300	300

5.3 Rough controls and short panels

A further recorded scalar comparison fixes three controls before the run: $g(Z) = 0.8(|Z| - \sqrt{2/\pi})$, $0.8 \text{sign}(Z)$, and $0.8\sqrt{2} \sin(3Z)$. These give a kink, a jump and rapid oscillation in 2,100 datasets (Table 3). Extrapolation’s worst null rate is 0.147, versus 0.177 for splines and 0.097 for KCI. No regime meets the joint criterion of lower null rejection than splines, disjoint pointwise 95% intervals and power within 0.05.

The panel study uses 80 entities, $M = 25, 50, 100$ independent periods, five whole-period folds, and 300 paired draws per data-generating setting. The independent-score null is $Y = a_i + 0.6B + 0.3 \sin(B) + \eta$, with independent normal B, X, η and fixed $a_i \in [-0.6, 0.6]$. A high-overlap null sets $X = B + 0.1\epsilon$, $Y = B + 0.1\eta$, where each period’s B is a permuted fixed 80-point grid plus a common shift. Given a focal baseline rank, the peer grid is fixed; exchangeable independent error arrays retain the conditional rank null. This argument does not cover arbitrary random peer grids. The positive control uses $Y = B + 0.1(0.6\epsilon + 0.8\eta)$.

At $M = 100$, $\beta = 2$ reduces the high-overlap shift: mean T changes from -13.52 to -0.30 in the released summaries; Table 4 gives rejection counts. Its independent-null rate is $11/300$, with a 95% Wilson interval $[0.021, 0.064]$ containing 0.05. At $M = 25$, however, both $\beta = 2$ null intervals lie above 0.05. At $M = 25$, for $\beta = 1$ in the independent-null case, the reported standard error is only 0.856 times the empirical standard deviation of the estimated mean. Both approximation bias and too few clusters can distort the test.

A local spline GCM uses all 2,700 simulated panels with identical ranks, controls, folds and cluster scales. Training quantiles define eight cubic-spline knots; duplicate knots are removed, and ridge penalty 0.001 applies only to spline coefficients, with unpenalized entity effects. At $M = 25$, its two null rates are $24/300$ and $30/300$, whose intervals also lie entirely above 0.05. At $M = 100$, both are $10/300$. All three methods reject every positive control, so these alternatives cannot establish a power advantage.

On all 41 public forecasts, splines use identical predictions, samples, folds, HAC lags and complete-family BY. All three guarded retained sets agree; Table 1 gives the extrapolation counts and the package gives the spline counts. Before guarding, splines reject the electricity copy ($T = 4.94$) and ratings copy ($T = 3.20$).

All 33 settings have 300 replications: 27 combinations of three conditions, three cluster counts and $\beta \in \{0.5, 1, 2\}$ on the primary ladder, plus six settings at $M = 100$, $\beta = 1$ using $\{4, 6, 8, 12, 16\}$ or $\{16, 24, 32, 48, 64\}$. At $M = 100$ and $\beta = 1$, the coarse ladder rejects $0/300$ positive-control panels, whereas the primary $\{8, 12, 16, 24, 32\}$ and fine ladders each reject $300/300$.

The smooth scalar study uses Fig. 2’s functions, known uniform bins and bivariate standard normal errors with correlation zero or 0.2. Three sample sizes and 300 paired draws give 5,400 datasets. Twofold histogram fits on $q_{\min}\{1, 2, 4\}$ compare coarsest, finest, and both extrapolated statistics: 21,600 calculations. With $q_{\min} = \lceil n^{1/5} \rceil$ and $\beta = 2$, linear and quadratic null rates range from 0.030 to 0.073. For the sine example they are 0.157, 0.087, and 0.120 at $n = 2,000, 8,000$, and $32,000$.

Population calculations confirm a substantial remainder despite the correct leading exponent.

5.4 Public-data forecast protocol

Four public datasets support five tasks: electricity and retail demand, ratings, return forecasts and factor alpha. Model families and temporal rules precede fitting. Panel audits use the five-resolution ladder with both exponents; the presentation order follows the panel checks. Ranks retain ties. Five contiguous blocks exclude whole evaluation clusters from each joint fit of additive entity and baseline-bin effects. Undefined ranks, negligible residuals or standard error at most 10^{-10} in standardized-rank product units yield $p^* = 1$, as do guarded copies. The floor prevents studentizing solver noise. Every model remains in BY; ten-bin adjustment and interleaved folds are sensitivities.

UCI Electricity [10] uses the first 48 meters in the verified provider archive, excluding incomplete boundary days. Eleven models forecast next-day mean demand with annual cutoffs in 2013 and 2014: 34,992 predictions per model over 729 days. The baseline is seven-day seasonal persistence; ranks are daily, with 14 HAC lags. Two-meter perturbation tests find no future-target influence on earlier predictions for all eleven models.

For M5 [11], a training-only rule selects 4,000 product-store series on a complete weekly grid. There are 992,000 training and 112,000 evaluation series-weeks, the latter spanning 28 complete weeks. The final partial week is excluded. Price features use previous weeks; normalization and imputation use training data. Histories update after each week, so the task is rolling one-week prediction. Ten models share last week’s sales as baseline, within-week ranks, and two weekly HAC lags. The small cluster count limits inference.

MovieLens-25M [12] uses ratings before January 1, 2017 for training and 2017 for evaluation. Training activity alone selects 6,000 users; 1,368,078 training ratings support ten models with 12,088 held-out ratings from 279 users. Items are entities, users are clusters, and the baseline is the training item mean. Within-user ranks assign zero to 18 singleton users. There is no HAC ordering across users; their clustering does not account automatically for shared-item dependence.

For Open Source Asset Pricing (OSAP) [13], all 173,302 long-short returns match the authors’ public 2025.10 release. Ten models use annual training cutoffs with monthly history updates. The target is the next calendar month’s return; the baseline compounds the preceding 12 months through the forecast origin. Evaluation covers 27,272 portfolio-months, 209 portfolios and 132 origins from January 2013 to December 2023, with targets February 2013 to January 2024. Three published portfolios have no admissible forecast rows. Ranks and folds are monthly, with 12 HAC lags. This evaluates published return series.

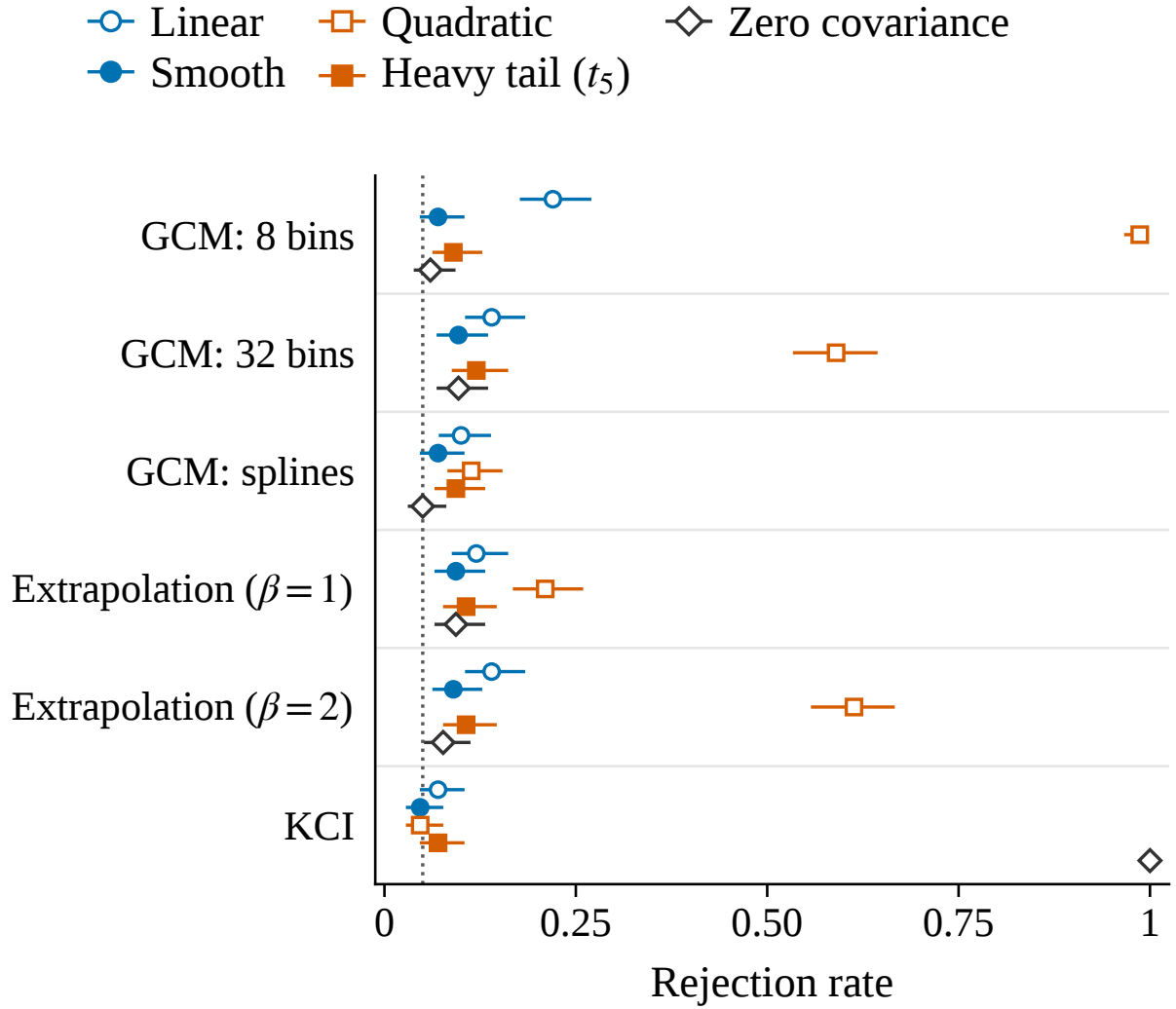
A separate OSAP task tests each long-short series’ intercept against zero in a Fama and French five-factor regression with momentum [6, 7]. Calendar-based Newey and West covariance [1] uses 12 monthly lags and two-sided normal-tail p -values. BY uses all 212 signals; six- and 24-lag estimates are sensitivities. This shares a data source with the forecast task, so it is not a fifth independent dataset.

5.5 Retrospective audit of a restricted case

We also audit stored predictions from thirty models in a UK insider-trading study. The data remain private. Each retained record has an observed next-calendar-month outcome and all twelve preceding calendar-month returns, excluding the forecast month, for the momentum control. This corrects incomplete histories and next-observed-month substitutions; the rule was fixed before recomputation after seeing historical results. Supports differ: 2,613 to 31,045 firm-months over 72 or 132 months. Available stored predictions are averaged within firm-month; source counts and target rounding differ, so this is not a common-sample model ranking.

This audit retains five whole-month folds, additive firm and momentum-bin effects, the same ladder, $\beta = 1$ primary and $\beta = 2$ sensitivity. Equation (1) uses months without HAC or a copy guard, differing from the public protocol. Historical full-panel imputation makes this a retrospective output audit. Only model-level aggregates are distributed.

A separate common-support reconstruction retrains the adversarial residual-ranking MLP, Ridge and histogram gradient boosting with training-only preprocessing and matured labels. Across all 55 fits, 18,036 observations and 132 common months, mean monthly Spearman correlations are MLP 0.00275, Ridge 0.05667, boosting 0.03794 and one-month momentum 0.02314. The MLP has no descriptive advantage; no significance comparison is claimed. The package records every seed aggregate and temporal rule.



Zero covariance is an alternative for KCI.

Figure 4: **Matched scalar comparison.** Each method shows the four independent nulls and the zero-covariance dependence case from Table 2; intervals are pointwise 95% Wilson intervals and the line marks 0.05. The dependence case is an alternative for KCI. All methods use the same data within a replication. The extrapolated statistic retains approximation error, particularly under the quadratic baseline.

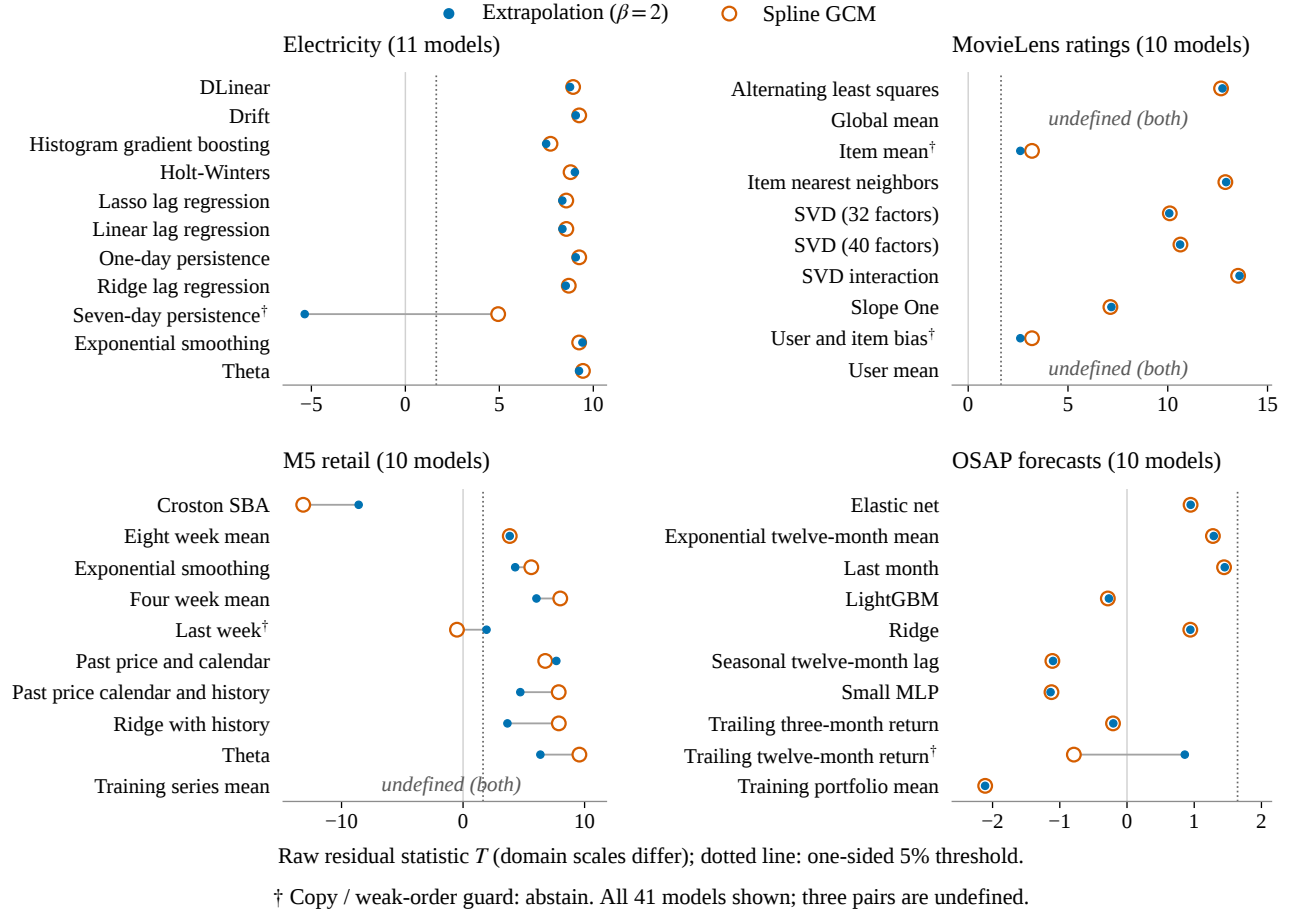


Figure 5: **Same forecasts, different adjustments.** All 41 public models, with 38 finite pairs and three undefined pairs. Points show raw extrapolated ($\beta = 2$) and spline statistics before the copy rule; the vertical line is the one-sided 5% normal threshold. Daggers identify finite copy or weak-order guards. Larger statistics do not establish better calibration or power. Full scores, MAEs and family-adjusted values accompany the reproduction package.

6 Estimated means, family inference and near-copy controls

6.1 Estimated conditional means for tied ranks

The main paper’s Corollary 1 gives a studentized limit with independent estimated means. The public generator `learned_reference_study.py` evaluates a finite-control instance. Let $Z \sim \text{Bernoulli}(1/2)$ and, conditionally independently, $V, W \mid Z \sim \text{Bernoulli}(0.2 + 0.6Z)$. Marginal midranks take values $1/4$ and $3/4$; their conditional means are $0.35 + 0.3Z$. The target is zero, $\Gamma = 9/400$, and the first-projection variance is $1/625$. These are exact values, checked by finite-state enumeration.

Training uses its own leave-one-out empirical midranks and two category means. It is independent of evaluation, but its fitted functions are shared across evaluation groups. Training sizes are 32, 512 and 8,192; exact means provide a diagnostic fourth arm. Group sizes are 16 and 64, with 25, 100 and 400 independent resampled groups. The 1,000 replications share evaluation primitives across arms and nested group sizes. There are 48 cells and 48,000 records. Population fitting-error products and L^2 bounds are evaluated from the known synthetic law; these diagnostics are not given to the test and are not confidence bounds available under an unknown law.

Across the six sampling configurations, distinct-reference rates are 0.041–0.055 with exact means, 0.136–0.527 with 32 training observations, 0.046–0.079 with 512, and 0.041–0.057 with 8,192. For 512 training observations, two of six pointwise Wilson intervals exclude 0.05, both with 400 groups. For 8,192, none does. The experiment supports the role of training accuracy, not a finite-sample universal threshold or a power comparison. The main captioned peer study retains the separate fixed-peer counterexamples.

6.2 Complete raw-score model families

The generator `independent_entity_family.py` tests the complete raw-score branch rather than only a marginal 0.05 rejection frequency. It uses $G = 25, 100, 400$ independent Gaussian entity trajectories, 20 observations, five contiguous folds and the same 12 middle observations per entity. Forecasts use 0.2 times the previous five outcome innovations plus unit-variance forecast noise. Forecast noises have pairwise correlation 0.5 across models, and all models share outcomes. Entity intercepts and all noises are independent across entities.

Families have 10 or 11 models. All-null families and families with three injected alternatives of coefficients 0.03, 0.06 or 0.10 share the same primitives. Each of 1,000 evaluation families runs the fitting, equal-entity studentization, numerical abstention and complete-family BY steps. Current-outcome injections are positive controls, not prospective forecasts. The full 216 cells retain failed methods and all hypotheses.

A separate 5,000-draw null bank is independent of evaluation. For each method and entity count, plus-one upper-tail ranks give Monte Carlo values with ties included. The same bank calibrates exchangeable model marginals. The bank’s minimum value $1/5001$ resolves the first BY thresholds for $K = 10, 11$. It supplies the known simulated null law, which is extra information. Reported evaluation-family Monte Carlo intervals are conditional on the realized bank; they do not integrate uncertainty over new banks. Pointwise Hoeffding intervals additionally bound finite-evaluation FDR uncertainty; a zero observed error rate does not imply zero risk. Normal, t_{G-1} and supplied-null references are all retained. The raw entity scale already uses sample standard deviation with divisor $G - 1$; applying a second degrees-of-freedom multiplier would be incorrect.

For $G = 400, K = 11$, the all-null FDR for complementary, gapped and directional fitting is 0, 0.928 and 0.011 with the normal reference. It is 0, 0.926 and 0.011 with t_{G-1} , and 0.008, 0.015 and 0.009 with independent supplied-null calibration. Across all six size/family combinations, directional normal-reference all-null FDR is 0.011–0.020. These simulations support the stated independent-entity branch, not a ranked, binned or serially dependent panel guarantee.

At coefficient 0.06, $G = 400, K = 11$, calibrated complementary and directional powers are 0.8443 and 0.6660. The paired difference, directional minus complementary, is -0.1783 , with pointwise Monte Carlo interval $[-0.1976, -0.1591]$. Both use independent null calibration at the same nominal level; their realized FDRs are not forced to be equal. Directional centering therefore provides an analytic alternative to an external null bank, not power dominance after supplying that additional information.

6.3 Near-copy representation and independent evaluation

The generator `near_copy_adjustment.py` uses 25 independent periods, 80 entities, a permuted fixed baseline grid and independent homogeneous Gaussian noises. Conditional on a focal baseline rank, the competing baseline multiset is fixed. The two full noise arrays are independent at zero correlation, so the two conditional rank distributions factorize. This supplies an exact nondegenerate rank null for positive score noise. Exact copies form a separate zero-product-variance boundary.

Comparators share five complementary folds, ranks, observations and the same adjustment information: 32 bins, both extrapolation exponents, eight-knot cubic splines, and splines with an explicitly unpenalized linear term. Exact conditional rank means are a diagnostic arm. The historical period-cluster statistic uses divisor M ; the small-sample variant uses $M - 1$ and t_{24} . Unlike the equal-entity study above, this changes the scale as well as the reference distribution.

For noise 0.01, 32 bins reject 984/1,000 evaluation null panels. The spline and linear-plus-spline normal counts are 57 and 54; their small-sample counts are 48 and 46. Exponent one rejects zero but has mean statistic -13.49 , so zero rejection alone is not evidence of centering. The new experiment is distinct from the historical fixed restricted-pattern stress, which gave 93/100 nominal positives for bins and 11/100 for splines and did not regenerate the full rank-null design.

Independent 1,000-draw calibration and 1,000-draw evaluation banks also compare weak injected correlations. The original 0, 0.3, 0.6 grid saturated its positive controls. The recorded exploratory extension adds 0.03, 0.06, 0.10 on the same primitives and retains every original cell. At noise 0.01 and correlation 0.06, supplied-null-calibrated powers are 0.125, 0.069, 0.157, 0.335, 0.344 and 0.338 for bins, exponent one, exponent two, splines, linear-plus-splines and exact means, respectively. The small linear-term difference is local and exploratory; it establishes no broad superiority over splines. All counts, paired differences and calibration inputs accompany the executable scripts.

6.4 A matched classical temporal moment

A further comparison uses the forward orthogonal deviation of Arellano and Bover [14], with a recursively centered predictable instrument. It compares a moment construction, not the complete panel GMM estimator. The method was specified after the preceding results were reviewed, with its formula and primary comparisons fixed before this additional run.

For the same 20 observed periods and evaluation rows $t = 5, \dots, 16$, set $c_t = \sqrt{(20 - t)/(21 - t)}$ and define

$$S_{gk}^{\text{FOD}} = \frac{\sum_{t=5}^{16} c_t (X_{gkt} - \bar{X}_{gk, < t})(Y_{gt} - \bar{Y}_{g, > t})}{\sum_{t=5}^{16} c_t}.$$

The instrument subtracts all preceding observed predictions, and the outcome subtracts all subsequent observed outcomes. Under the stated innovation conditions, the instrument is orthogonal to current and future outcome innovations, so this score has exact null mean zero. Independent entities permit the same sample-standard-deviation studentization. The positive design-only denominator leaves the statistic unchanged from averaging the weighted products, while making the mean equal to the constant innovation-injection coefficient τ . The blocked directional score has the same mean τ . Under time-varying effects, the two target averages have different weights; their equivalence is specific to the constant-effect comparison. More explicitly, replacing the constant injection by $X_t = X_t^0 + \tau_t \varepsilon_t$ with independent unit-variance innovations and fixed τ_t gives expected directional mean $12^{-1} \sum_{t=5}^{16} \tau_t$ and classical mean $\sum_{t=5}^{16} c_t \tau_t / \sum_{t=5}^{16} c_t$. The earlier-centered forecast contains no current or future innovation, except for its declared current injection, so expansion leaves only τ_t in each product. Unequal c_t establish the different weights algebraically; no efficiency advantage is inferred from this identity.

All original primitive draws, evaluation rows, entity weights and full $K = 10, 11$ families are reused. The new method receives its own 5,000 calibration statistics on the same independent null draws, followed by 1,000 evaluation families. Every original method's means, standard errors, statistics, probabilities and BY decisions replay identically. The four-method study has 288 cells, including all three reference laws, three entity counts and all four signal levels. It extends the preceding study; repeated cells are not independent confirmation.

At $G = 400, K = 11$, the classical moment's all-null FDR is 0.012 with the normal reference, 0.011 with t_{399} , and 0.015 with supplied-null calibration. The corresponding directional values are 0.011, 0.011 and 0.009. For $\tau = 0.06$, normal-reference power is 0.7517 versus 0.6577; calibrated power is 0.7523 versus 0.6660. The calibrated paired power difference is 0.0863, with pointwise Monte Carlo interval $[0.0721, 0.1006]$ and Bonferroni-adjusted normal Monte Carlo interval $[0.0637, 0.1089]$ over the 27 primary contrasts. The latter is an approximate simultaneous Monte Carlo interval, conditional on the calibration bank, not a finite-sample statistical guarantee. Pointwise Hoeffding FDR bounds accompany the normal and Wilson summaries.

The classical moment has higher power than the blocked directional score in this design. Calibrated complementary fitting is still more powerful (0.8443), but requires the externally supplied null law to correct its biased reference. These comparisons separate exact centering from statistical efficiency and do not establish power dominance for the proposed blocked construction. The generator, input compatibility checks, all saved arrays, paired intervals and literal-product tests are released.

7 Standard cluster GCM conditions

The next result applies the standard GCM argument at the cluster scale [2, 3].

Theorem 1 (Conditional validity). *Under (A1) to (A3) below, with nuisance approximation bias negligible relative to the standard error, the statistic T_{cl} in Eq. (1) converges in distribution to $\mathcal{N}(0, 1)$ under conditional independence. The one-sided level- α test is consistent for fixed alternatives with $\theta_\infty > 0$ and a standard error tending to zero.*

The proof also covers a pointwise zero target with oracle mean $o(a_n)$. For a fixed negative target, a consistent mean and vanishing standard error give rejection probability tending to zero. Neither statement is uniform over drifting nulls. Training excludes complete evaluation clusters.

Assumptions For the oracle product combination, let s_i^* use population nuisance regressions, $L_g^* = \sum_{i \in g} s_i^*$, and $\mathbb{E}[L_g^*] = n_g \mu_n$, treating cluster sizes as fixed. Define $v_n^2 = \sum_g \text{Var}(L_g^*)$ and $a_n = v_n/n$.

(A1) *Cluster sampling and central limit theorem.* The oracle cluster sums are independent across clusters, $M \rightarrow \infty$, $\max_g n_g/n \rightarrow 0$, and $a_n \rightarrow 0$. For some $\nu \in (0, 2]$, $\sum_g \mathbb{E}|L_g^* - n_g \mu_n|^{2+\nu}/v_n^{2+\nu} \rightarrow 0$.

(A2) *Negligible nuisance remainder.* With $\Delta_g = L_g - L_g^*$, require $\sum_g \Delta_g = o_p(v_n)$ and $\sum_g \Delta_g^2 = o_p(v_n^2)$. In an i.i.d. setting, the usual product-rate condition $\|\hat{g}_X - g_X\|_2 \|\hat{g}_Y - g_Y\|_2 = o_p(n^{-1/2})$ controls the quadratic remainder, while weighted prediction-error and moment bounds control the linear terms. Growing clusters require those bounds at scale a_n .

(A3) *Non-degenerate product scale.* $0 < v_n^2 < \infty$ and the cluster-product moments in (A1) exist. In the i.i.d. specialization invoking $\sqrt{n}$ rates, additionally assume $v_n^2/n \rightarrow \sigma^2 \in (0, \infty)$.

Relation to the implementation Nuisances refer to the declared score. Cross-fitting and HAC do not alone prove (A2) and (A1).

Proof of Theorem 1. Expanding fitted products gives $R - R^* = -\Delta_X e_Y^* - e_X^* \Delta_Y + \Delta_X \Delta_Y$, where $\Delta = \hat{g} - g$. Assumption (A2) controls these terms at the cluster scale. By (A1), the centered oracle sum divided by v_n tends to $\mathcal{N}(0, 1)$ and its sum of squared centered cluster contributions, divided by v_n^2 , tends in probability to one. Empirical centering changes this ratio by $o_p(1)$ because $\bar{s} - \mu_n = O_p(a_n)$ and $\sum_g n_g^2/n^2 \leq \max_g n_g/n \rightarrow 0$. Both parts of (A2) and the Cauchy-Schwarz inequality then permit fitted products. Thus $\hat{s}_{\text{cl}}/a_n \rightarrow_p 1$. A null mean $o(a_n)$ gives the normal limit; a positive limiting mean and vanishing scale give consistency.

8 Conditional resolution theory

The main paper records the fixed-grid score as a sensitivity analysis. This section retains the full conditional expansion and its counterexamples. Assumptions (A1)–(A3) are stated in the preceding section; (A4) is given below.

Additional assumption (A4) *Resolution bias expansion.* The population mean obeys $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ with a common c , a specified $\beta > 0$, and a remainder bound uniform over the ladder. For an expanding ladder take fixed m , $q_j/q_{\min} \rightarrow a_j$ with distinct positive a_j , so the intercept weights have bounded absolute sum. Inference also requires the stated bias-to-standard-error rate. This is an assumption on the residual rank covariance, not a consequence of assigning percentiles.

8.1 A conditional resolution correction

Assume the population numerator has the expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ with a specified exponent $\beta > 0$. On a resolution grid $\mathcal{Q} = \{q_1, \dots, q_m\}$ of distinct resolutions, let w be the intercept weights from regressing a vector of product means on $\{1, q^{-\beta}\}$. They satisfy $\sum_j w_j = 1$ and $\sum_j w_j q_j^{-\beta} = 0$. Define

$$s_i = \sum_{j=1}^m w_j R_i^{(q_j)}, \quad T_{\text{ex}} = \bar{s}/\hat{s}_{\text{e}}(s). \quad (4)$$

Product means are combined before studentization; their cross-resolution covariance enters the variance. Use Eq. (1) for independent clusters or the declared HAC scale for a screen.

Theorem 2 (Leading-bias cancellation). *Under (A4), with bounded $\sum_j |w_j|$, the population mean of the oracle combination s^* is $\theta_\infty + b_n$, where $b_n = O(q_{\min}^{-2\beta})$. If (A1) to (A3) hold for this combination and $b_n/a_n \rightarrow 0$, where a_n is its oracle standard error, then $T_{\text{ex}} \Rightarrow \mathcal{N}(0, 1)$ under conditional independence. For an i.i.d. non-degenerate scale, a sufficient bias condition is $\sqrt{n} q_{\min}^{-2\beta} \rightarrow 0$. For fixed $\theta_\infty > 0$, $b_n \rightarrow 0$ and $a_n \rightarrow 0$ give consistency.*

The following calculations derive $\beta = 2$ for uniform scalar controls with C^3 means and the rate window $n^{1/8} \ll q_{\min} \leq q_{\max} \ll n^{1/2}$. Gaussian quantile bins instead have bias of order $1/(q \log q)$; bounded ranks can also violate the q^{-2} expansion (Proposition 2). Lemma 1 quantifies exponent error. We consequently treat fixed-grid extrapolation as a conditional correction and report both exponents alongside splines.

Proof of leading-bias cancellation. For extrapolation set $u_j = q_j^{-\beta}$ and $X_Q = [\mathbf{1} \ u]$. The intercept weights are

$$w = X_Q (X_Q^\top X_Q)^{-1} (1, 0)^\top.$$

They have $\mathbf{1}^\top w = 1$, $u^\top w = 0$ and bounded absolute sum. The combined mean is therefore $\theta_\infty + O(q_{\min}^{-2\beta})$. Applying the main paper's GCM limit to this combination gives the claimed limit when $b_n/a_n \rightarrow 0$. At a fixed ladder, the admissible null expansion $cq^{-\beta} + dq^{-2\beta}$ instead leaves $d \sum_j w_j q_j^{-2\beta}$, generally nonzero. Separate per-resolution validity does not supply the combination's assumptions.

Proposition 1 (A smooth scalar case). *Let $U \sim \text{Uniform}[0, 1]$, $X = f(U) + \epsilon$, $Y = g(U) + \eta$, with square-integrable conditionally mean-zero errors and $f, g \in C^3[0, 1]$. For the population means f_q, g_q on q equal-width bins,*

$$\begin{aligned} \mathbb{E}[(X - f_q(U))(Y - g_q(U))] \\ = \theta + \frac{1}{12q^2} \int_0^1 f'(u)g'(u) du + O(q^{-4}), \end{aligned} \quad (5)$$

where $\theta = \mathbb{E}[\epsilon\eta]$. Thus (A4) holds with $\beta = 2$ in this setting.

Proof. Write $h = 1/q$ and expand about each bin midpoint c . After subtracting bin means, the leading terms are $f'(c)V + \frac{1}{2}f''(c)(V^2 - \mathbb{E}V^2)$ and the corresponding expression for g , with a uniform $O(h^3)$ remainder and $V \sim \text{Uniform}[-h/2, h/2]$. Odd moments vanish, giving bin covariance $f'(c)g'(c)h^2/12 + O(h^4)$. The midpoint approximation to the integral of $f'g'$ has error $O(h^2)$. Averaging the bin covariances proves the expansion.

For the rate window, use i.i.d. observations, known uniform bins, deterministic proportional twofold splits and training-fold histogram means. Assume bounded conditional fourth moments and $0 < \text{Var}(\epsilon\eta) < \infty$. Histogram fitting error is $O_p(\sqrt{q_{\max}/n})$ in L^2 . The linear terms are conditionally centered; the quadratic mean is $O_p(q_{\max}/n)$. Thus $q_{\max} = o(\sqrt{n})$ makes fitting error negligible at root- n scale, while bounded fourth moments give variance consistency. Combining this with $\sqrt{n}q_{\min}^{-4} \rightarrow 0$ gives the stated nonempty window. The sine results show finite-sample error despite this rate window; this calculation does not establish the panel assumptions.

Gaussian quantile bins need not have the assumed leading power This failure already occurs with oracle adjustment, without fitting error. Let $Z \sim \mathcal{N}(0, 1)$, let $I_q(Z)$ index its $q \geq 2$ equal-probability bins, and write $z_j = \Phi^{-1}(j/q)$, with $\phi(\pm\infty) = 0$. For $X = aZ + \epsilon$, $Y = aZ + \eta$, with Z, ϵ, η mutually independent and both errors square-integrable and mean-zero, the binned residual covariance is $a^2 D_q$, where

$$D_q = 1 - q \sum_{j=1}^q [\phi(z_{j-1}) - \phi(z_j)]^2 \asymp \frac{1}{q \log q}. \quad (6)$$

Thus neither $c/q + O(q^{-2})$ nor $c/q^2 + O(q^{-4})$ holds for $a \neq 0$. This is an application of scalar quantization [15], not a new quantization principle. To prove the order, put $t = z_{q-1}$, $\lambda = \phi(t)/(1 - \Phi(t))$ and $v_t = 1 + t\lambda - \lambda^2$. The two tail bins contribute $2v_t/q$. Interior conditional variances are at most one quarter of bin width squared; the Cauchy-Schwarz inequality therefore gives

$$\frac{2v_t}{q} \leq D_q \leq \frac{2v_t}{q} + \frac{1}{4q^2} \int_{-t}^t \frac{dz}{\phi(z)}. \quad (7)$$

The normal-tail expansion gives $v_t \sim t^{-2}$, $t^2 \sim 2 \log q$, $\phi(t) \sim t/q$, and the integral is asymptotic to $2/(t\phi(t))$. Hence $1 \leq \liminf q \log(q) D_q \leq \limsup q \log(q) D_q \leq 5/4$.

Proposition 2 (A boundary obstruction for population ranks). *Let $Z, \epsilon_X, \epsilon_Y$ be independent standard normals and $X = \rho_X Z + \sqrt{1 - \rho_X^2} \epsilon_X$, $Y = \rho_Y Z + \sqrt{1 - \rho_Y^2} \epsilon_Y$, with $0 < \rho_X, \rho_Y < 1$. Set $V = \Phi(X)$, $W = \Phi(Y)$, $U = \Phi(Z)$, $a = \rho_X / \sqrt{2 - \rho_X^2}$ and $b = \rho_Y / \sqrt{2 - \rho_Y^2}$. For the oracle residual covariance B_q after q equal-width bins of U ,*

$$a^2 + b^2 \leq 1 \implies q^2 B_q \longrightarrow \infty. \quad (8)$$

Thus bounded population ranks do not guarantee a finite leading q^{-2} coefficient, even though $X \perp Y \mid Z$.

Proof. The conditional means are $f(u) = \Phi(a\Phi^{-1}(u))$ and $g(u) = \Phi(b\Phi^{-1}(u))$, obtained by integrating over an independent standard normal. Every bin covariance is nonnegative because both means increase. On any fixed $[\delta, 1 - \delta]$, their derivatives are bounded; the midpoint expansion, retaining only fully interior bins, gives $\liminf q^2 B_q \geq \frac{1}{12} \int_{\delta}^{1-\delta} f' g' dz$. With $z = \Phi^{-1}(u)$,

$$f'(u)g'(u) du = \frac{ab}{\sqrt{2\pi}} e^{-(a^2+b^2-1)z^2/2} dz. \quad (9)$$

This integral diverges as $\delta \downarrow 0$ when $a^2 + b^2 \leq 1$, proving the claim. For equal correlations the condition is $\rho_X = \rho_Y \leq \sqrt{2/3}$. A finite derivative integral in the other regime alone does not establish the $O(q^{-4})$ remainder in (A4).

Misspecified exponents For distinct $q_1, \dots, q_m > 0$, $m \geq 2$, let $w(\beta)$ be the ordinary least-squares (OLS) intercept weights on $[1, q^{-\beta}]$, and $q_+ = \max_j q_j$. Define the remaining leading bias in finest-resolution units by

$$\varrho(\beta, \beta_\star) = q_+^{\beta_\star} \sum_j w_j(\beta) q_j^{-\beta_\star}. \quad (10)$$

Lemma 1 (Direction of exponent error). *For positive exponents, $\varrho = 0$ exactly when $\beta = \beta_\star$; otherwise $\text{sign}(\varrho) = \text{sign}(\beta - \beta_\star)$. The positive ratio need not be at most one.*

Proof. Put $u_j = q_j^{-\beta}$ and $p = \beta_\star / \beta$. The intercept of u^p has positive denominator $m \sum_j u_j^2 - (\sum_j u_j)^2$ and numerator

$$\sum_{i < j} u_i u_j (u_j - u_i) (u_i^{p-1} - u_j^{p-1}). \quad (11)$$

Every term has the stated sign; all vanish when $p = 1$.

On $\{8, 12, 16, 24, 32\}$, $\varrho(2, 0.5) = 1.066$ and $\varrho(2, 1) = 1.046$: remaining leading bias can exceed that of the finest bin.

For $\mu(q) = \theta + cq^{-\beta_\star} + r(q)$, the intercept bias is $cq_+^{-\beta_\star} \varrho + \sum_j w_j r(q_j)$. Underestimating the exponent makes only the leading term conservative for a positive one-sided test when $c > 0$; for $c < 0$ the sign reverses. The smooth scalar coefficient is $c = \int f' g' / 12$, not necessarily positive. Remainder, nuisance and sampling errors still require control. With $\Sigma = \text{Cov}((\tilde{R}^{(q_j)})_{j=1}^m)$, the intercept variance is $w^\top \Sigma w$; $\|w\|_2^2$ compares variance only when $\Sigma \propto I$. Estimated tied ranks, raw-scale smoothness or Lipschitz continuity alone do not establish the required signed expansion.

9 Observable category budgets and a general covariance boundary

The main paper's reference theorem uses all three observation roles in the order-three U-statistic. A concrete dependent boundary has constant controls, V uniform on $\{0, 1, 2, 3\}$ and $W = (1, 3, 0, 2)_V$. The focal residual product takes $(3, -3, -3, 3)/64$; each reference-role term takes $(1, -1, -1, 1)/64$. Hence the target and reuse coefficient are zero, but the full variance is $25/4096$, compared with focal variance $9/4096$. This is a counterexample to using focal variance under an arbitrary covariance null, not an error in the released group denominator. Across 3,000 independent trials at $M = 400$, $N = 64$, the group test rejects 143 times and the deliberately focal-only test 482 times. Exact count enumeration, generic triple kernels and the full projection are independently checked in the release. For a local mixture with $W = V$ of weight $c/\sqrt{MN}$, the target is $(5/64)c/\sqrt{MN}$ and the limiting statistic is $\mathcal{N}(c, 1)$. The local sequence keeps the full projection variance away from zero.

The initial absolute-range categorical certificate replaces an externally supplied error allowance with an observable validation construction. Each validation pair contains an independent focal and reference row; both observations are charged. Conditional-mean confidence rectangles and simultaneous binomial probability limits share the failure budget $\delta = 10^{-4}$. The small linear program in the main paper accounts for rare and unobserved cells. This uses classical Hoeffding and binomial confidence arguments; those inequalities are not new contributions.

The study has two or eight categories, balanced or with one probability 0.005. Given category c , both Bernoulli margins have success probability p_c and covariance $\gamma p_c(1 - p_c)$. The two-category probabilities are (0.2, 0.8); the eight probabilities are equally spaced from 0.2 to 0.8. The midrank target is $\gamma \sum_c \Pr(Z = c) p_c(1 - p_c)/4$. Training sizes are 32, 512, 8,192; validation uses 512 or 8,192 pairs; $M = 100$ or 400, $N = 64$, and $\gamma = 0, 0.25, 0.5$. Each of 108 configurations has 1,000 replications. Every original maximum-cell result and its probability-weighted follow-up is retained. The weighting design followed the original rare-cell failure and reused its random primitives; it is not a new independent confirmation.

Table 5: Finite-certificate positive decisions out of 1,000. All use 8,192 training observations, 8,192 validation pairs and 400 groups of 64 peers: 50,176 total observations.

Category law	γ	Target	Maximum cell	Probability weighted
Two balanced	0	0	0	0
Two balanced	0.25	0.01000	65	54
Two balanced	0.50	0.02000	1,000	1,000
Eight balanced	0	0	0	0
Eight balanced	0.25	0.01321	0	0
Eight balanced	0.50	0.02643	1,000	1,000
Eight, rare cell	0	0	0	0
Eight, rare cell	0.25	0.01366	0	0
Eight, rare cell	0.50	0.02731	0	1,000

No nuisance-envelope or lower-bound coverage failure occurred in the recorded grids. This is not a zero-risk claim. Weak-signal power remains limited. Because weighting reallocates the failure budget, it need not improve each realization: 30 paired decisions favor only maximum-cell, whereas 3,182 favor only weighting. Generic covariance-null theory and fixed-family BY validity are mathematical results; this experiment measures single-certificate behavior rather than a new full-family benchmark.

10 Exact decomposition of the retail fit changes

For the unchanged 68,000 middle-fold rows over 17 weeks, define complementary residuals r_X, r_Y and directional-minus-complementary fitted mean changes Δ_X, Δ_Y . At each resolution, $\Delta \bar{R} = -r_X \Delta_Y - r_Y \Delta_X + \Delta_X \Delta_Y$. Combining each term with the unchanged exponent-two intercept weights preserves this identity. All three terms reinforce each change at

Table 6: Empirical mean-score change, decomposed by fitted-mean shifts.

Model	Outcome fit	Forecast fit	Joint term	Total
Price/calendar	-0.00381653	-0.00448701	-0.00341526	-0.01171880
Croston SBA	0.01242305	0.00970073	0.00327125	0.02539503

every tested resolution. The largest contribution is the forecast-mean shift for price/calendar and the outcome-mean shift for Croston. The full 15×15 cross-resolution/term HAC covariance is released; original means and standard errors reproduce. These are empirical fitting components, not identified causal shares of the linear population feedback model.

11 Locked public rental-demand confirmation

The Seoul Bike Sharing Demand data [16] contain 8,760 hourly observations from December 2017 through November 2018. The provider’s unchanged source and attribution are included under CC BY 4.0. The protocol was frozen locally before download, and models, adjustments, coefficients and choices were frozen before confirmation outcomes were parsed. This is a historical reanalysis of a newly obtained source, not an external registration, third-party replication or new transport domain for the project.

Seven-day embargoes separate the stages. Inputs are past counts at lags 1, 2, 3, 24, 25, 48, 168, past-only daily/weekly means and calendar harmonics. No contemporary weather, operational status or outcome innovation is used. The previous hourly count is assumed available; publication latency and historical provider vintages are not reconstructed. Confirmation outcomes enter only later lag features in sequential one-hour forecasting, never a model refit or selection. Perturbing all current/future counts leaves all forecasts at or before each checked origin unchanged.

Purpose	Inclusive dates	Hours
Forecast training	2017-12-08 to 2018-04-27	3,384
Nuisance fitting	2018-05-05 to 2018-06-29	1,344
Audit and selection	2018-07-07 to 2018-08-31	1,344
Confirmation	2018-09-08 to 2018-11-30	2,016

Table 7: Locked Seoul confirmation: raw rental-count MSE. All choices were fixed before the final 12 weeks. A gate admits only BY-retained candidates.

Forecast rule	Seasonal control	Ridge control
Adjusted baseline	333,914	53,554
Conventional convex blend	40,231	36,747
Ungated residual augmentation	34,125	31,123
Audit-gated augmentation	333,914	31,123

Two baseline controls, previous-day counts and Ridge, each have eight candidates: previous hour/day/week, Ridge, histogram gradient boosting, reflected Ridge, an exact baseline copy and a noise-perturbed baseline. The adjustment uses an intercept, linear baseline term and five-knot cubic splines, fitted only in the nuisance window. These are estimated projections, not true conditional means. Conventional convex blending, ungated residual augmentation and BY-gated augmentation each select by the audit-window MSE, with the adjusted baseline always available. The 16 pairs remain in the family. The gate uses eight weekly means, lag-one Bartlett HAC, an 8/7 correction and nominal t_7 values; none of these choices proves real-data calibration.

On Ridge, gated and ungated augmentation are identical. Their confirmation gain over the adjusted baseline is 22,431.43 squared counts. Their gain over conventional blending is 5,623.72, but weekly and fortnight intervals give different evidence about its sign. The direct boosting forecast has MSE 40,231.11; its difference from the augmentation is inconclusive under both references. On the seasonal baseline, gating blocks the useful augmentation and loses 299,789.52 squared counts relative to ungated selection. Both branches remain; none was chosen after confirmation. Intervals are pointwise temporal approximations, not exact or simultaneous coverage. All 74 risk rows, 82 paired comparisons, 16 gain identities and 888 weekly aggregates are released.

The exact empirical gain of fixed fits and coefficient t is $2t\hat{\theta}_{\text{confirm}} - t^2\hat{v}_{\text{confirm}}$. For Ridge, $t = 0.786989$, whereas the ex-post empirical optimum is 0.942380. The difference in squared coefficients explains 626.95 squared counts relative to that empirical optimum. It is neither the unknown population optimum nor a coefficient used for prediction. The projected gain identity measures what the learned combination achieved; it does not establish a general benefit from gating.

12 Rectangular validation: sufficient sample cost

Use the rectangular allowances defined in the main paper, with its range R , independent-triple count J , and separate validation and evaluation error allocations. This result is retained for that construction only. For the rectangular choice $\hat{B} = \hat{B}_+$, the observable width $W_B = \hat{B}_+ - \hat{B}_-$ supports sample planning. On its validation event, the range-only bound has conditional power at least $1 - \eta$, $0 < \eta < 1$, if

$$\vartheta > W_B + \frac{R}{\sqrt{2J}} \left(\sqrt{\log \frac{1}{\alpha - \delta}} + \sqrt{\log \frac{1}{\eta}} \right). \quad (12)$$

This sufficient condition uses the joint rectangular event, not the aggregate bound combined with the old lower allowance. It is not an optimal detection boundary. Count training and both members of every validation pair in the total budget.

Proof. On the rectangular validation event, $b_H \geq \hat{B}_-$ and $\mathbb{E}\bar{D} = \vartheta + b_H$. Conditional on validation, Hoeffding’s lower-tail bound gives

$$\bar{D} \geq \vartheta + b_H - R\sqrt{\log(1/\eta)/(2J)}$$

with probability at least $1 - \eta$. Subtract $\hat{B}_+$ and the range-only radius. The resulting lower bound is positive under Eq. (12). The rectangular event jointly controls both allowances; an independently constructed aggregate upper bound and this lower allowance require the appropriate union allocation.

13 Signed allowances, variance and observation cost

The refined reference certificate in the main paper separates a signed upper learning-bias limit from a sampling allowance. The original absolute-range construction and every previously released result remain available. This section specifies the new comparisons and their limits.

Signed interval optimization For each validation confidence rectangle, evaluating the bilinear product at all four corners gives its exact minimum and maximum. The probability constraints are unchanged from the absolute construction. An exchange argument proves the greedy optimizer: moving available mass from a smaller coefficient to a larger one cannot decrease the objective. Negative coefficients still receive mass when needed to satisfy total mass one. For identical intervals and caps, each signed upper coefficient is no larger than its absolute-error product, hence $\hat{B}_+ \leq \hat{B}_{\text{abs}}$ pointwise. No additional confidence failure is incurred. This comparison does not license changing the validation size or error allocation after viewing the evaluation results.

For a concrete persistent-error example, take one category and Bernoulli margins of $1/2$ with joint probabilities $P_{00} = P_{11} = 0.29$, $P_{01} = P_{10} = 0.21$. Marginal midranks are $1/4$ and $3/4$, so $\vartheta = 0.01$. Fixed fits $f = 0.65$, $g = 0.35$ give $b_H = -0.0225$. As validation and evaluation sizes grow, the absolute lower bound tends to -0.035 while the signed bound tends to 0.01 . The point is loss from subtracting an unnecessary absolute value under persistent misspecification, not a new generic concentration inequality or superiority over classical signed inference.

Variance from independent triples Partition evaluation rows by their predeclared order within each group. For each consecutive triple, average all six permutations of focal, forecast-reference and outcome-reference roles. Ignore the final $N \bmod 3$ rows only for this variance estimate; the full statistic retains them. The J symmetric kernels are independent under the common-law design. Their ordinary sample variance uses denominator $J - 1$. The proof in the main paper applies the variance confidence bound and the full-U concentration event by a union bound; it does not assume independence between this sample variance and the full mean. The factor two in the inverted variance-based probability is essential. Taking the minimum of separately valid range and variance probabilities without an error allocation would be a different, unsupported procedure.

The Bernstein constant For the latter constant, the Bernstein bound is $\log \mathbb{E} e^{\lambda(\bar{D} - \mathbb{E}[\bar{D}|H])} \leq v\lambda^2/[2(1 - c\lambda)]$, with $v = \sigma^2/J$, $c = R/(3J)$. Setting $t = \sqrt{2x/v}$ and $\lambda = t/(1 + ct)$ gives exponent x at $\sqrt{2vx} + cx$; the zero-variance case follows directly.

Comparators The independent-triple comparator uses the *mean of these triples*, with radius

$$s\sqrt{2x/J} + \frac{7Rx}{3(J-1)}, \quad x = \log\{2/(\alpha - \delta)\},$$

from the classical empirical Bernstein bound [17]. Applying this radius directly to the full U-statistic is not justified by that theorem. The pooled comparator computes the full order-three U-statistic over all MN peers and measures its kernel variance with $\lfloor MN/3 \rfloor$ triples. Pooling is legitimate here because every group follows the same law. Both comparators receive the same signed learning allowance, fitted means, validation pairs and evaluation rows. They are strong alternatives, not methods assumed to have lower power.

Fixed simulation grid The two-category law has equal category masses and Bernoulli success probabilities $(0.2, 0.8)$. The eight-category law has masses $(0.005, 0.995/7, \dots, 0.995/7)$ and success probabilities equally spaced from 0.2 to 0.8 . Conditional covariance is $\gamma p_c(1 - p_c)$, with $\gamma \in \{-0.125, 0, 0.125, 0.25, 0.5\}$. The target is $\gamma \sum_c \Pr(Z = c)p_c(1 - p_c)/4$. Training uses 8,192 observations and empirical-midrank cell fits. Validation has 512 or 8,192 disjoint pairs; evaluation has $M \in \{25, 100, 400, 1600\}$ groups of $N = 64$ peers. The two-category 8,192-pair setting additionally shifts the fitted means by $+0.15$ and -0.15 , clipped to $[0, 1]$. Five constructions give 500 cells with 1,000 paired replications per cell, at $\alpha = 0.05$, $\delta = 10^{-4}$. These are single-certificate comparisons. A separate helper chooses a predeclared family-size-aware validation budget; it does not retune this study's error allocation or turn it into a new family experiment.

The study protocol and executable hashes were saved before the full run. A two-replication computational pilot checked the same declared grid. After adding a defensive check to the independent-triple API, the entire 500,000-row computation was repeated and both result files were byte-identical. This is a reproducibility check, not a second statistical confirmation. The public protocol preserves the original and replay links.

Table 8: Positive decisions out of 1,000, with estimated means and 8,192 validation pairs. Every row counts all training and evaluation observations and both members of each validation pair.

Law	ϑ	Total rows	Absolute range	Signed range	Variance full U	Triples Bernstein	Pooled variance
Two	0.010	30,976	0	0	1,000	1,000	1,000
Two	0.010	50,176	58	78	1,000	1,000	1,000
Two	0.005	30,976	0	0	6	24	6
Two	0.005	50,176	0	0	1,000	1,000	1,000
Eight	0.01366	30,976	0	0	435	433	461
Eight	0.01366	50,176	0	0	1,000	1,000	1,000
Eight	0.00683	50,176	0	0	0	0	0

All 200 nonpositive cells have zero observed rejections, and all 500 cells have zero observed lower-bound coverage failures. Neither means zero risk; for a single 0/1000 binomial cell the one-sided 95% upper limit is approximately 0.003. All 512-pair validation settings remain undetected. Rare categories and weak signals therefore still expose a learning-budget bottleneck. The new full-U variance bound adds 11,565 positive decisions relative to the absolute-range construction across the paired grid, but the benefit is largely shared by classical comparators. Relative to independent triples, its net gain is only 38 decisions; relative to pooled variance it has 51 fewer. These grid sums are descriptive, not population-weighted performance measures.

Sample planning and family resolution For a desired effect $\vartheta_0 > W_B$, solving the main paper’s sufficient power condition gives a strict lower requirement on the number of triples:

$$J > \frac{R^2}{2(\vartheta_0 - W_B)^2} \left(\sqrt{\log \frac{1}{\alpha - \delta}} + \sqrt{\log \frac{1}{\eta}} \right)^2.$$

Choose evaluation size after independent validation and before evaluating new rows. The guarantee is conditional on the simultaneous validation event, whose failure probability is at most δ . No finite count is certified by this condition when $\vartheta_0 \leq W_B$; collecting only more evaluation rows cannot remove that sufficient-condition bottleneck. This is not a necessary lower bound or an optimal allocation theorem. For a fixed K -model BY family, setting $\delta = \rho\alpha/(KH_K)$ with $0 < \rho < 1$ keeps the probability floor below the first-rank threshold. With fixed $\delta = 10^{-4}$ and $K = 100$, that threshold is about 9.64×10^{-5} . One isolated strong signal cannot pass the first rank, although later step-up ranks can still reject.

14 A finite-population certificate on a public archive

The source is UCI ElectricityLoadDiagrams20112014 [10], under CC BY 4.0. We keep the first 96 source meters and nominal daily means of the provider’s 96 quarter-hour readings. A single histogram gradient boosting model trains on 67,392 rows before 2013 and remains fixed over 730 days in 2013–2014. Lagged observations update only as they become available. The source’s zeros are retained, including those before some meters became active. Categories combine target weekend with whether the previous-week forecast exceeds its pre-2013 training median. That frozen median is zero. Category counts in the 70,080-row archive are 883, 49,229, 350 and 19,618.

Conditional on the fixed archive, independent replacement draws in the training, validation and evaluation streams have a common finite-population law. Repeated row identities do not create dependence between independently drawn indices. This establishes the sampling assumptions literally for the finite-archive target, irrespective of temporal dependence in the source. It does not establish independence of chronological observations or inference about a future population. The source had previously been used in the project; this is not an unseen source or external preregistration.

The three candidates are boosting, the previous-week forecast and a category-code copy. The protocol fixes all seven method comparisons, 4,096 mean-fitting rows, 8,192 validation pairs and 12,288 evaluation rows before computing the audit. The primary pooled-U variance certificate has boosting lower bound 0.070503 and seasonal lower bound 0.069418. Exact census targets, computed after writing every certificate, are 0.078219 and 0.077414. All seven methods retain both forecasts; the category-copy control is not retained. The common BY-adjusted values for the two positives are 0.000275.

The 32,768 queries access 26,140 unique rows. Forecast preparation already accesses the complete source archive and takes 8.39 seconds in the recorded environment. Census targets for all three candidates take about 0.039 seconds, versus 0.328 seconds for one pooled boosting U-statistic; its independent triples take about 0.00046 seconds. Thus no end-to-end access or runtime advantage is established. The coarse categories leave substantial continuous-baseline information, as the positive

seasonal candidate illustrates. The result demonstrates a valid finite-archive certificate, not additional forecast value beyond the full seasonal baseline. The legal derived archive, draw indices, frozen model, protocol and all bounds are included. A portable source-data replay reproduces the forecasts and numerical results within the recorded numerical tolerances; timings naturally vary.

15 Why the rental-demand gate blocked a useful candidate

The seasonal-control boosting candidate was not flagged as a copy. Its raw-count residual mean was 125,102.410, with lag-one weekly HAC standard error 38,057.967 over eight selection weeks, giving $T = 3.28715$ and nominal t_7 probability 0.00667738. It ranked third in the complete 16-pair family, whose third threshold is 0.00277307. The monotone BY-adjusted value is 0.10940387, attained via the next rank's 0.0080903 probability. The eight weekly means are all positive but uneven. The HAC standard error is only 3.41% above the independent-week standard error; limited weekly replication and the family threshold are the main numerical bottlenecks. No rank transformation is used in this study.

The boosting candidate under Ridge control has raw probability 0.0000344919 and adjusted probability 0.00186572, and is selected by both gated and ungated rules. Four useful seasonal alternatives are blocked; two harmful Ridge alternatives are also blocked but would not be selected by the ungated rule. Hence the gate provides no realized Ridge benefit and incurs 299,789.52 MSE opportunity loss under the seasonal control. Removing guarded hypotheses, splitting the family or replacing the tail law after observing this result would create a different procedure. Such arithmetic sensitivities are labelled retrospective in the release; none is presented as a newly confirmed gate.

16 Additional finite bound and weighting identities

16.1 A finite-sample certificate with a bias allowance

For the finite guarantee, fit nuisances on independent training clusters. Let H contain the training and design information. For evaluation cluster g , let S_g be its mean combined residual product. Prespecified nonnegative weights λ_g sum to one. The target $\theta_H = \sum_g \lambda_g \mathbb{E}[\langle X_g - m_{Xg}, Y_g - m_{Yg} \rangle / n_g \mid H]$ is the weighted oracle residual covariance, where $m_{Xg} = \mathbb{E}[X_g \mid C_g, H]$, m_{Yg} likewise, and C_g is the full evaluation-control array of cluster g .

Proposition 3 (Bias-aware finite certificate). *Conditional on H , suppose the evaluation clusters are independent and $S_g \in [\ell_g, u_g]$ almost surely. All weights and bounds are H -measurable. Write $\mu_H = \sum_g \lambda_g \mathbb{E}[S_g \mid H]$ and $V_H = \sum_g \lambda_g^2 (u_g - \ell_g)^2$. If an H -measurable allowance B satisfies $\Pr(\mu_H - \theta_H \leq B) \geq 1 - \delta$, for $0 \leq \delta < \alpha$, then*

$$L_\alpha = \sum_g \lambda_g S_g - B - \sqrt{\frac{1}{2} V_H \log(1/(\alpha - \delta))} \quad (13)$$

satisfies $\Pr(\theta_H \geq L_\alpha) \geq 1 - \alpha$. Thus deciding positively only when $L_\alpha > 0$ has type-I error at most α under $\theta_H \leq 0$ almost surely.

Proof. Conditional Hoeffding concentration [18] bounds the upper tail of $\sum_g \lambda_g S_g - \mu_H$ by $\exp(-2x^2/V_H)$. Allocate $\alpha - \delta$ to this tail and δ to failure of the bias allowance, then apply a union bound. If $V_H = 0$, the weighted score equals its conditional mean almost surely.

With $\lambda_g = n_g/n$ the target is pooled-row; equal weights give an average-cluster target. A flat resolution curve or a good held-out fit does not supply B , and total prediction risk in place of conditional-mean error gives a valid but powerless allowance. The real-data normal screens do not apply this finite theorem.

The weighting tradeoff Bias cancellation can increase variance. With $u_j = q_j^{-\beta}$ and $X_Q = [\mathbf{1} \ u]$, the classical constrained optimum for $\Sigma \succ 0$ is $w_G = \Sigma^{-1} X_Q (X_Q^\top \Sigma^{-1} X_Q)^{-1} (1, 0)^\top$. For any feasible w , $X_Q^\top (w - w_G) = 0$, so

$$w^\top \Sigma w = w_G^\top \Sigma w_G + (w - w_G)^\top \Sigma (w - w_G). \quad (14)$$

An additional second-order constraint costs variance strictly only if $(u^{\odot 2})^\top w_G \neq 0$. On our ladder at $\beta = 1$, it raises the minimum squared weight norm by a factor of 6.44, a variance ratio only for $\Sigma \propto I$. We use fixed intercept weights.

The cost of a second bias constraint This calculation is classical constrained least squares, consistent with generalized bias reduction [19]. For $m \geq 3$ distinct positive resolutions and fixed $\beta > 0$, write $u_j = q_j^{-\beta}$, $h_j = u_j^2$, $C = [\mathbf{1} \ u]$, $w_0 = C(C^\top C)^{-1}(1, 0)^\top$, and $r = [I - C(C^\top C)^{-1}C^\top]h$. Let $\kappa_0 = h^\top w_0$. Among weights satisfying $C^\top w = (1, 0)^\top$ and $h^\top w = \kappa$, the unique minimum-norm solution and its squared norm are

$$\begin{aligned} w_*(\kappa) &= w_0 + \frac{\kappa - \kappa_0}{r^\top r} r, \\ V(\kappa) &= \|w_0\|^2 + \frac{(\kappa - \kappa_0)^2}{r^\top r}. \end{aligned} \quad (15)$$

Indeed, feasible perturbations are orthogonal to w_0 ; their projection onto r supplies the additional constraint. For $\mathcal{Q} = \{8, 12, 16, 24, 32\}$ and $\beta = 1$, $\kappa_0 = -5845/1179648$, $r^\top r = 7777/1811939328$, $V(\kappa_0) = 269/256$, and $V(0) = 22567/3333$. Thus full second-order cancellation raises the minimum squared norm by $V(0)/V(\kappa_0) = 6.44356$; the endpoint directions have cosine 0.39395. This is a variance ratio only for isotropic cross-resolution covariance. For general $\Sigma \succ 0$, the constrained GLS optimum is $w_G = \Sigma^{-1}C(C^\top \Sigma^{-1}C)^{-1}(1, 0)^\top$; adding $h^\top w = 0$ costs variance strictly only if $h^\top w_G \neq 0$. Neither result implies a false-positive floor. Changing β changes the constraint class and is not this fixed- β frontier. For the intercept weights $w(\beta)$ on $[1, q^{-\beta}]$, define $\varrho(\beta, \beta_*) = q_+^{\beta_*} \sum_j w_j q_j^{-\beta_*}$. The preceding resolution analysis proves its sign is that of $\beta - \beta_*$ when nonzero.

Table 9: Remaining leading bias in finest-resolution units on $\{8, 12, 16, 24, 32\}$. Rows give the assumed exponent; columns give the true exponent. Values are deterministic, rounded to three decimals.

β	β_*					$\ w\ _2$
	0.5	1	1.5	2	3	
0.5	0	-2.119	-6.483	-15.336	-68.823	1.874
1	0.714	0	-1.611	-5.074	-27.228	1.025
1.5	0.950	0.700	0	-1.679	-13.468	0.774
2	1.066	1.046	0.796	0	-6.656	0.664

16.2 Separating rank-reference bias from nuisance error

Ranks introduce a second bias when both channels reuse the same finite reference sample. To isolate it, condition on independent training data H and take $N \geq 3$ i.i.d. triples (V_i, W_i, Z_i) per independent group, with a common law. Let F_V, F_W be population marginal mid-distributions, $m_V(Z) = \mathbb{E}[F_V(V) \mid Z, H]$, and m_W analogous. This result targets $\vartheta = \mathbb{E}[(F_V(V) - m_V)(F_W(W) - m_W) \mid H]$, which differs from the standardized panel-rank target. Set $a_{ij} = \mathbf{1}\{V_j < V_i\} + \frac{1}{2}\mathbf{1}\{V_j = V_i\}$, define b_{ij} similarly, and let $k = N - 1$, $U_i = k^{-1} \sum_{j \neq i} a_{ij}$, $V'_i = k^{-1} \sum_{j \neq i} b_{ij}$. For H -measurable nuisance fits $f, g \in [0, 1]$, define

$$\begin{aligned} C_i &= (U_i - f(Z_i))(V'_i - g(Z_i)), \\ D_i &= C_i + \frac{U_i V'_i - k^{-1} \sum_{j \neq i} a_{ij} b_{ij}}{k - 1}. \end{aligned} \quad (16)$$

Proposition 4 (Reference-reuse decomposition). *Conditional on H , write $\Gamma = \mathbb{E}[a_{12}b_{12} - F_V(V_1)F_W(W_1) \mid H]$ and $b_H = \mathbb{E}[(m_V - f)(m_W - g) \mid H]$. Then, including ties,*

$$\begin{aligned} \mathbb{E}[C_i \mid H] &= \vartheta + b_H + \frac{\Gamma}{N - 1}, \\ \mathbb{E}[D_i \mid H] &= \vartheta + b_H. \end{aligned} \quad (17)$$

Distinct reference indices remove the $\Gamma/(N - 1)$ term exactly; either exact nuisance removes b_H . The following argument gives the proof and a finite confidence bound using classical U-statistic concentration [4, 18]. Independent groups alone are insufficient: dependent peers can leave positive bias even with exact nuisances. This correction is therefore a separate construction under i.i.d. peer sampling, not a repair for arbitrary dependent panels.

Distinct-reference proof and bound For distinct i, j, l , reference independence conditional on the focal row and H gives $\mathbb{E}[a_{ij}b_{il} \mid V_i, W_i, Z_i, H] = F_V(V_i)F_W(W_i)$. The score D_i averages $(a_{ij} - f(Z_i))(b_{il} - g(Z_i))$ over these ordered pairs.

Conditional centering removes both linear nuisance-error terms, leaving $\vartheta + b_H$. The $N - 1$ shared pairs among $(N - 1)^2$ pairs in C_i add $\Gamma/(N - 1)$. This proves Eq. (17), including ties.

For M equal independent groups, let $\bar{D}$ average their group means. With an H -measurable $B \geq b_H$ almost surely,

$$L_\alpha = \bar{D} - B - \sqrt{\frac{2\log(1/\alpha)}{M\lfloor N/3 \rfloor}} \quad (18)$$

has conditional coverage at least $1 - \alpha$ for ϑ . Indeed, the symmetrized degree-three kernel lies in $[-1, 1]$. Averaging over permutations into disjoint triples recovers each group mean. Exponential convexity and Hoeffding's lemma bound its centered MGF by $\exp\{t^2/(2\lfloor N/3 \rfloor)\}$; group independence gives the stated radius. If the training envelope fails with probability δ , replace α by $\alpha - \delta > 0$ for unconditional coverage.

Group independence alone is insufficient. Let Z be a fair bit, $V = Z \text{ xor } E_V$, $W = Z \text{ xor } E_W$, with independent flips of probabilities $1/4$ and $1/3$. Duplicate one triple within each group. Then $\vartheta = 0$, and exact nuisances are $m_V(z) = 3/8 + z/4$, $m_W(z) = 5/12 + z/6$. All comparisons equal $1/2$, giving $D_i = (1/2 - m_V(Z))(1/2 - m_W(Z)) = 1/96 > 0$. Thus dependent peers invalidate the reference argument even with exact fits.

Theorem 3 (Oracle sampling-ratio consequence). *Consider $M \rightarrow \infty$ independent groups, each containing $N \rightarrow \infty$ i.i.d. rows from the same fixed law. Suppose $V \perp W \mid Z$ and both nuisances in Eq. (16) are exact: $f = m_V$, $g = m_W$. Set*

$$r_V = F_V(V) - m_V(Z), \quad r_W = F_W(W) - m_W(Z), \quad \sigma^2 = \mathbb{E}[r_V^2 r_W^2] > 0.$$

Let C_g, D_g be the group averages of C_i, D_i , respectively. For $A = C, D$, write $\bar{A} = M^{-1} \sum_g A_g$ and $\hat{v}_A = (M - 1)^{-1} \sum_g (A_g - \bar{A})^2$. Define $T_A = \sqrt{M} \bar{A} / \sqrt{\hat{v}_A}$ for $A = C, D$. Then

$$\begin{aligned} \sqrt{MN} \bar{D} &\Rightarrow \mathcal{N}(0, \sigma^2), \\ \sqrt{MN} \{\bar{C} - \Gamma/(N - 1)\} &\Rightarrow \mathcal{N}(0, \sigma^2), \\ N \hat{v}_D &\rightarrow_p \sigma^2, \quad N \hat{v}_C \rightarrow_p \sigma^2. \end{aligned} \quad (19)$$

Thus T_D converges to $\mathcal{N}(0, 1)$. If $M/N \rightarrow c \in [0, \infty)$, then $T_C \Rightarrow \mathcal{N}(\Gamma\sqrt{c}/\sigma, 1)$. If $\Gamma > 0$ and $M/N \rightarrow \infty$, then $T_C \rightarrow_p +\infty$. The limiting one-sided naive rejection probability is therefore $1 - \Phi(z_{1-\alpha} - \Gamma\sqrt{c}/\sigma)$ for finite c , and one in the divergent positive-bias case.

Proof. Symmetrize the ordered-triple kernel $(a_{ij} - m_V(Z_i))(b_{il} - m_W(Z_i))$ over its six role assignments. Under the raw conditional null its first Hoeffding projection is $h_1(O) = r_V r_W / 3$. The focal role contributes $r_V r_W$. For the reference- V role, the focal W residual has conditional mean zero given its controls and focal V , so that role contributes zero. The reference- W role is analogous. The U-statistic decomposition gives

$$D_g = N^{-1} \sum_i r_{V,gi} r_{W,gi} + R_g, \quad \mathbb{E} R_g = 0, \quad \mathbb{E} R_g^2 = O(N^{-2}).$$

Group independence makes the variance of $\sqrt{MN} \bar{R}$ of order N^{-1} . The MN leading bounded i.i.d. terms obey the ordinary CLT.

Let J_g average $a_{ij} b_{ij}$ over ordered pairs $i \neq j$, and let Q_g average $a_{ij} b_{il}$ over ordered distinct triples. The definitions give $C_g - D_g = (J_g - Q_g)/(N - 1)$ and $\mathbb{E}(J_g - Q_g) = \Gamma$. Bounded kernels of degrees two and three imply $\text{Var}(J_g - Q_g) = O(N^{-1})$. Consequently the variance of

$$\frac{\sqrt{MN}}{N - 1} \left\{ M^{-1} \sum_g (J_g - Q_g) - \Gamma \right\}$$

is $O(N^{-2})$, proving the centered shared-reference CLT.

For either statistic, the centered fourth moment is $O(N^{-2})$. To see this, represent the bounded U-statistic as an average over permutations of disjoint triples, then apply convexity to the fourth power. Each permutation average contains $\lfloor N/3 \rfloor$ independent bounded centered terms. The shared-reference statistic has the same bound by embedding its degree-two component in a degree-three kernel with uniformly bounded coefficients. Hence the variance of the scaled cluster second-moment average is $O(M^{-1})$; replacing its true mean by the sample mean costs $O_p(M^{-1})$ after multiplication by N . Together with the variance limits from the decompositions above, this proves consistency of $N \hat{v}_A$.

Slutsky's theorem gives the finite- c limits. In the divergent case, $T_C / [\Gamma\sqrt{MN}/\{(N - 1)\sigma\}] \rightarrow_p 1$. This ratio argument does not require a centered normal approximation at a diverging studentized shift.

Both exact conditional means and nonzero first-projection variance are essential to this statement. Honest training alone is insufficient. It gives no exact small- M calibration and does not apply to dependent or within-period standardized panel ranks. More independent groups can make reference reuse detectable even when the population residual covariance is zero; distinct references remove that specific error.

17 A comparison with supplied null simulations

The main scalar comparison uses each method’s normal or gamma reference. Their different null rejection rates preclude a direct power ranking. Here we ask a narrower question: how do the six fixed statistics compare when each receives an independent bank from the exact same null generator? This is a retrospective secondary analysis of existing outputs. The split, all methods and all comparisons were fixed before the secondary calculation; they were not fixed before the original simulation results became available.

For each of the four scalar regimes, replications 0 through 149 under the null form a calibration bank of $B = 150$ fully observed datasets, each with $n = 400$. Null replications 150 through 299 and all 300 positive-covariance replications are evaluation data. The generating seeds and sample hashes are disjoint across these roles. All six methods receive the same observations within each replication. The extra zero-covariance dependence design is excluded because it has different null membership for KCI and covariance tests.

Write p_b for the native reference value on bank dataset b and p_0 for that on an evaluation dataset. Use

$$p_{MC} = \frac{1 + \sum_{b=1}^{150} \mathbf{1}\{p_b \leq p_0\}}{151}. \quad (20)$$

Under exchangeable bank and new-null datasets and a fixed score function, the rank argument gives $\Pr(p_{MC} \leq 0.05) \leq 7/151 = 0.04636$. To see this, an exchangeable continuous score has a uniform rank among 151 values; the inclusive comparison makes ties conservative. This is a classical Monte Carlo reference [20], not a new learned calibration method. It requires a supplied null generating law and does not apply to the observational panels in the main paper.

The guarantee averages over the bank as well as the evaluation dataset. After one bank is fixed, its conditional rejection probability can differ from $7/151$. The 150 validation indicators are independent conditional on that bank, but share calibration randomness marginally. Accordingly, all Wilson intervals in Fig. 6 describe the conditional rate for the realized bank; they do not validate the marginal rank theorem.

Table 10: Oracle-bank comparison. Each entry is the rejection count divided by its denominator: 150 validation nulls or 300 positive-covariance alternatives. The same bank is used within each method and regime.

Method	Linear		Smooth		Quadratic		Heavy tails	
	Null	Power	Null	Power	Null	Power	Null	Power
GCM, 8 bins	7/150	278/300	7/150	292/300	4/150	179/300	3/150	289/300
GCM, 32 bins	12/150	274/300	9/150	282/300	6/150	241/300	6/150	278/300
GCM, splines	8/150	275/300	6/150	280/300	8/150	280/300	4/150	279/300
Extrapolation, $\beta = 1$	11/150	273/300	10/150	281/300	12/150	256/300	6/150	278/300
Extrapolation, $\beta = 2$	11/150	283/300	10/150	287/300	8/150	246/300	5/150	286/300
KCI, gamma	13/150	218/300	11/150	226/300	4/150	231/300	6/150	268/300

At exponent one, extrapolation detects 273/300 and 281/300 linear and smooth alternatives, compared with KCI’s 218/300 and 226/300. In the quadratic design, splines detect 280/300, compared with 256/300 for exponent one and 246/300 for exponent two. All 15 method pairs in all four positive-effect regimes are compared with two-sided exact discordance tests, followed by Holm correction over 60 comparisons. The two exponent-one gains over KCI and the quadratic spline gain over exponent one remain significant, with adjusted values 2.53×10^{-11} , 4.93×10^{-10} and 0.00765. These comparisons are conditional on the realized calibration banks. All 10,800 evaluated method rows, 48 cells and 60 contrasts are retained.

The result identifies a conditional power tradeoff under supplied null information; it does not identify a uniformly preferable adjustment. The cost includes 150 complete oracle null datasets per regime, and the secondary analysis supplies no evidence of actual-panel validity or independent confirmation on new observations.

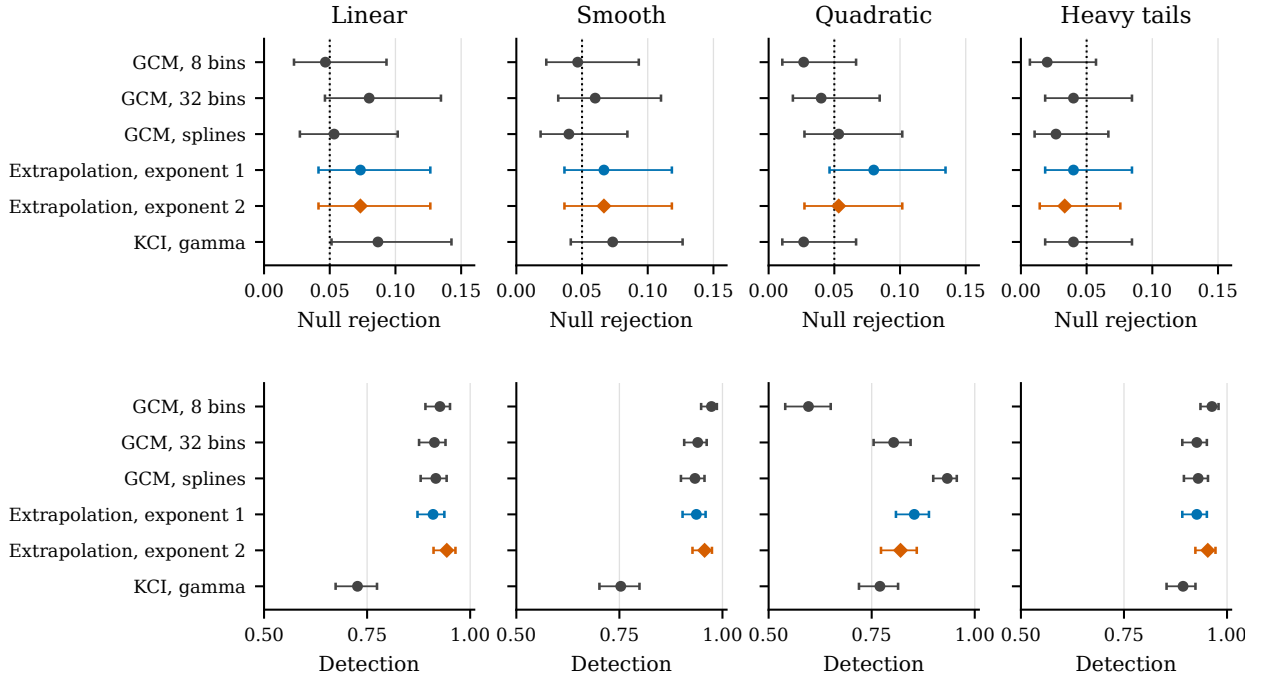


Figure 6: **Calibration and detection with a supplied null bank.** All 48 method, regime and effect cells are shown. Points and intervals are the observed rates and conditional-bank 95% Wilson intervals. The dotted line in the null panels marks 0.05. Colors distinguish the two extrapolation exponents; all methods share the same displayed scales. The calibration datasets are additional oracle information, unavailable for direct use on the real panels.

18 Legacy sensitivity experiments and comparator scope

We reproduce four earlier simulation drivers with their original settings, retaining every decision and every unfavorable result. These experiments use empirical within-period ranks against random peers and row-random nuisance folds. Their zero latent added signal does not establish conditional independence of the ranked quantities given the focal baseline rank and entity. We therefore report rejection under zero or positive latent added signal, rather than calibrated Type-I error or size-corrected power. The separate matched-null studies in the main paper address a different design.

The synthetic additive nuisance fit jointly estimates firm, period and momentum-bin effects within each training fold. It is distinct from the historical sequential real-panel adjustment. Every simulated panel is balanced, so the clustered SE's mean-total centering equals size centering. These two implementation checks do not establish independent folds, negligible nuisance error or valid ranked-panel normal-reference inference.

18.1 Adaptive exponent and complete ablation

The adaptive heuristic fits a log slope to absolute successive differences of the five evaluated product means, clips the exponent to $[0.5, 2]$, and tests weights selected using those same means. It omits the varying adjacent-resolution factor in a pure-power difference and treats the selected weights as fixed in its SE. A noiseless spectrum $\theta + cq^{-1}$ already yields $\hat{\beta} = 1.180886$ on this ladder. In contrast, the geometric triple 8, 16, 32 identifies a positive pure-power order for $c \neq 0$ from the ratio of successive differences. Failure of this heuristic does not prove that the order cannot be estimated.

Table 11 gives all four regimes, with fifty paired draws per signal condition. The average estimated order is computed on the zero-signal draws. Independent draws within each cell give the pointwise Wilson intervals in Figure 7; they are Monte Carlo intervals, not guarantees for the scientific null.

Table 11: Adaptive-exponent replay. Entries are rejected draws/total draws. Zero and positive denote latent signal coefficients 0 and 0.15.

Regime	Zero signal		Positive signal		Mean $\hat{\beta}$
	Fixed	Adaptive	Fixed	Adaptive	
Smooth	3/50	4/50	50/50	50/50	1.303450
Nonlinear	7/50	48/50	50/50	50/50	1.960841
Larger panel	5/50	8/50	50/50	50/50	1.834468
Heavy tails	5/50	4/50	48/50	47/50	1.109617

Changing the assumed order changes the leading-bias constraint. On the five-resolution ladder, squared weight norms fall from 1.050781 at $\beta = 1$ to 0.449668 at 1.96 and 0.441348 at 2. At 2 the weights no longer annihilate the q^{-1} term. This is not the fixed-order second-order-annihilation frontier; the observed rejection change does not identify a unique causal mechanism.

The complete ablation uses overlap 0.8, four regimes and nine settings. Each signal cell has 150 draws except the larger panel's 80; the two signal conditions use disjoint seed ranges. All settings share the same data draws within each condition. There is no fold-scheme ablation. The short ladder is $\{8, 16, 32\}$ and the long ladder is $\{8, 12, 16, 20, 24, 28, 32, 40\}$. Other settings use $\{8, 12, 16, 24, 32\}$, equal-mass bins, twelve fitting sweeps and exponent one unless the row changes that component. Table 12 retains the long ladder as well as all exponent changes.

Table 12: Complete ablation. Each cell gives zero-signal / positive-signal rejection rates. Larger-panel rates use 80 draws; other rates use 150.

Setting	Smooth	Nonlinear	Larger panel	Heavy tails
Default	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933
Short ladder	0.127 / 1.000	0.160 / 0.987	0.100 / 1.000	0.087 / 0.933
Long ladder	0.113 / 1.000	0.220 / 0.987	0.113 / 1.000	0.093 / 0.960
$\beta = 0.5$	0.047 / 0.960	0.000 / 0.000	0.050 / 1.000	0.033 / 0.847
$\beta = 1.5$	0.147 / 1.000	0.833 / 1.000	0.125 / 1.000	0.147 / 0.960
$\beta = 2$	0.180 / 1.000	0.973 / 1.000	0.163 / 1.000	0.147 / 0.987
Equal-width bins	0.113 / 1.000	0.153 / 0.987	0.113 / 1.000	0.087 / 0.940
4 fit sweeps	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933
24 fit sweeps	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933

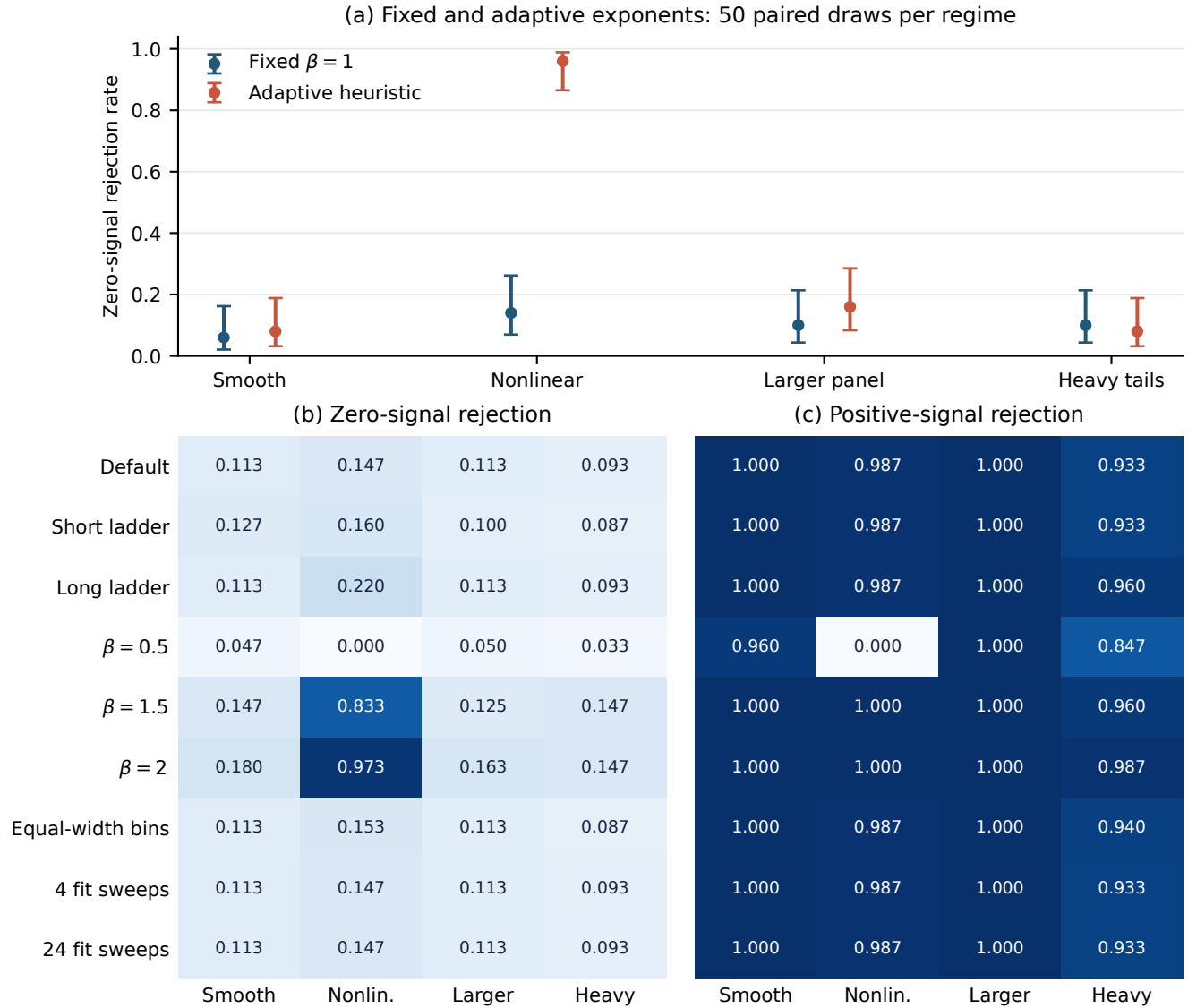


Figure 7: Complete legacy sensitivity, with no settings dropped. Panel (a) shows pointwise 95% Wilson intervals over 50 draws per method/regime. Panels (b,c) show all ablation rates, with 150 draws except 80 in the larger panel. These random-peer-ranked experiments do not validate the ranked conditional null. Full integer counts, denominators and cellwise intervals are supplied in the machine-readable tables.

18.2 Historical grid and stress designs

The earlier grid has eight implementations, five overlaps $\{0.5, 0.65, 0.8, 0.9, 0.95\}$ and four regimes: 160 method/design pairs, each with two signal conditions. Each condition uses 150 draws, except 80 for the larger panel. The legacy criterion requires zero-signal rejection at most 0.18 and positive-signal rejection at least 0.90; it is a descriptive tolerance rule, not nominal 0.05 calibration. A stricter historical prose criterion used less than 0.15 and more than 0.95. Table 13 makes the two criteria explicit.

The comparator names describe the implemented variants. Coarse/fine GCM use IID product SEs, while extrapolation uses month clustering. The linear variant fits a second-stage rank-momentum slope on the full sample. Sign weighting and binned projection learn from residuals that already depend on outer evaluation labels; deterministic target perturbations confirm this feedback. Local permutation conditions on month and momentum bins but omits firm. The grid therefore does not establish performance of the published WGCM, PCM or local-permutation procedures under their respective assumptions. The separately mislabeled RFF/distance routines are absent from this eight-implementation grid.

Table 13: Legacy grid summary over twenty designs per implementation. Extrema are descriptive; failure counts refer only to the stated tolerance rules and do not measure scientific-null validity.

Implementation	Maximum	Minimum	Failed designs/20	
	Zero signal	Positive signal	.18/.90	.15/.95
Cluster extrapolation	0.147	0.953	0/20	0/20
Coarse GCM	1.000	0.993	20/20	20/20
Fine GCM	1.000	0.973	16/20	19/20
Rank-linear projection	1.000	1.000	20/20	20/20
Naive permutation	1.000	0.993	20/20	20/20
Legacy sign weighting	0.920	0.507	15/20	20/20
Legacy binned projection	1.000	0.520	18/20	20/20
Month/bin permutation	0.633	0.847	10/20	15/20

Stress designs use 600 firms/60 periods, 120/100 with serial latent processes, 120/98 after a three-period rolling outcome sum, or 120/24. The rolling sum includes the current period, so it is not a strict forward forecasting target. A common additive period shock disappears under within-period ranking; these constructed processes do not isolate a unique cause of miscalibration. All positive-signal losses remain in Table 14.

Table 14: All stress conditions. Rates with pointwise 95% Wilson Monte Carlo intervals; these are not simultaneous intervals or ranked-null guarantees.

Design	Draws	Zero signal	Positive signal
Many fixed effects	100	0.030 [0.010, 0.085]	1.000 [0.963, 1.000]
Serial processes	200	0.085 [0.054, 0.132]	0.990 [0.964, 0.997]
Overlapping outcomes	200	0.105 [0.070, 0.155]	0.750 [0.686, 0.805]
Twenty-four periods	200	0.175 [0.129, 0.234]	0.760 [0.696, 0.814]

These replays recover reproducible sensitivity measurements without restoring claims of an unavoidable error floor, unique sensitivity to the exponent, or superiority over faithful native comparator implementations. The accompanying machine-readable tables retain all 416 rate cells, integer counts, exact denominators and pointwise intervals, including all 320 grid rates. No settings are changed after observing the replay.

19 What an exact reference can establish

The following benchmark separates three questions: whether an estimator preserves its covariance target, whether its null distribution is valid, and whether learning improves detection. Its reference distribution is classical Gaussian regression geometry [21, 22]. It does not establish a new distribution-free certificate for the ranked observational panels in the main paper.

19.1 Preserving the target before testing it

Condition on full controls and independent training information. Let clusters be independent, with arbitrary dependence within each cluster. Write $x = X - h_X$, assume $E[Y] = \Phi\gamma$ for a declared full-column-rank design Φ , and allow $E[x]$ to be unrestricted. The fixed-design target is

$$\theta = \sum_g \text{tr}\{D_g \text{Cov}(Y_g, X_g)\}, \quad D_g = I_{n_g}/n. \quad (21)$$

Here $n = \sum_g n_g$. Equal cluster weights would define a different target.

Proposition 5 (Block preservation and feasibility). *A design-only bilinear estimator $x^\top AY$ is unbiased for Eq. (21), for every score mean, γ and feasible within-cluster cross-covariance, if and only if $A\Phi = 0$ and $A_{gg} = D_g$ for every cluster. Such a matrix exists if and only if deleting each cluster preserves the column rank of Φ .*

Proof. Its expectation is $E[x]^\top A\Phi\gamma + \sum_g \text{tr}\{A_{gg} \text{Cov}(Y_g, X_g)\}$. Varying the score mean, γ and each cross-covariance gives both necessary conditions; substitution gives sufficiency. The row-block equation is $A_{g,-g}\Phi_{-g} = -D_g\Phi_g$. If $\Phi_{-g}v = 0$, invertibility of D_g forces $\Phi_g v = 0$, hence $v = 0$. Conversely, full rank permits

$$A_{gg} = D_g, \quad A_{g,-g} = -D_g\Phi_g(\Phi_{-g}^\top\Phi_{-g})^{-1}\Phi_{-g}^\top. \quad (22)$$

This is the cluster leave-out regression construction of established leave-out estimation [23], specialized to the covariance target. A period-specific dummy loses all support when its period is deleted, so unrestricted period means fail the criterion. A pseudoinverse does not remove that obstruction. Cancelling only $\Phi^\top A\Phi$ or normalizing only $\text{tr}(A)$ is insufficient. For example, with $\Phi = [\mathbf{1}, (-1, 0, 2)^\top]$, $v = (2, -3, 1)^\top$ and $A = vv^\top/14$, both $\Phi^\top A\Phi = 0$ and $\text{tr}(A) = 1$ hold. Independent rows with cross-covariances $(-1, 1, 0)$ have average covariance zero but $E[X^\top AY] = 5/14$.

19.2 The role of a learned score mean

Under full-channel null independence, let $\delta = E[X] - h_X$, $\Sigma_X = \text{Var}(X)$, $\Sigma_Y = \text{Var}(Y)$ and $K = A\Sigma_Y A^\top$. Since $A\Phi = 0$, conditioning first on X gives

$$\text{Var}\{(X - h_X)^\top AY\} = \text{tr}(K\Sigma_X) + \delta^\top K\delta. \quad (23)$$

Learning helps this variance when it reduces the second term. Ordinary prediction mean squared error need not order errors in the metric K . If $E[Y] = \Phi\gamma + r$, the remaining bias is exactly $\delta^\top Ar$. Neither independent training nor a small training loss certifies that unobserved remainder. Empirical Gram corrections also have direct higher-order influence-function predecessors [24]; we do not claim that replacing a power law with a Gram matrix is itself new.

19.3 A finite reference with known covariance shape

Now use a separate model, which may include serial dependence:

$$Y \mid X, Z, H \sim \mathcal{N}(\Phi\gamma, \sigma^2 V), \quad V = LL^\top \succ 0. \quad (24)$$

The mean span and covariance shape V are supplied correctly before evaluation outcomes; only γ and the positive scale σ are unknown. H denotes independent training. Let P project orthogonally away from $L^{-1}\Phi$, let $d = \text{rank}(P) \geq 2$, and set $r = PL^{-1}Y$. For an outcome-independent raw direction $a = a(X, Z, H)$, put $u = PL^\top a / \|PL^\top a\|$ when this direction is nonzero.

Proposition 6 (Classical Gaussian reference). *Under Eq. (24),*

$$T_a = \frac{u^\top r}{\sqrt{\{\|r\|^2 - (u^\top r)^2\}/(d-1)}} \sim t_{d-1}. \quad (25)$$

Proof. In an orthonormal basis of the residual space whose first vector is u , the first Gaussian coordinate is independent of the other $d-1$ coordinates. Their squared norm, divided by σ^2 , is chi-squared with $d-1$ degrees of freedom. The nuisance mean has been projected out.

For Eq. (22), use $a = A^\top(X - h_X)$. The identity $\Phi^\top a = 0$ means the reference projection does not alter its numerator. Shared leave-out training otherwise makes cluster summands dependent; their sample standard deviation is not an exact reference. Known serial V validates Eq. (25), but it does not extend the independent-cluster covariance-target identity.

An estimated or incorrect V , omitted mean directions, arbitrary non-Gaussian errors and directions selected using evaluation outcomes are not covered. Directly ranking Y does not preserve this Gaussian model. The null is the stated conditional no-additional-mean model, not every distribution with zero signed covariance. These assumptions are true by construction in the following simulation, not verified for real forecasts.

19.4 A finite comparison with the classical regression direction

For $Y = \Phi\gamma + \tau X + \sigma L\epsilon$, $\tau \geq 0$, let $s = PL^{-1}X \neq 0$. The generalized least-squares (GLS) direction is $u_* = s/\|s\|$. Among tests of the form Eq. (25) with the same Φ, V and one-sided level $\alpha < 1/2$, this direction has at least as much conditional power as any other outcome-independent direction.

To prove this, a general unit direction u gives independent numerator and denominator components

$$N_u \sim \mathcal{N}(\tau u^\top s / \sigma, 1), \quad D_u \sim \chi_{d-1}^2(\tau^2\{\|s\|^2 - (u^\top s)^2\}/\sigma^2). \quad (26)$$

The aligned direction maximizes the numerator mean by Cauchy-Schwarz and sets denominator noncentrality to zero. A noncentral chi-squared variable is stochastically larger than its central counterpart, as seen from its Poisson mixture. The positive critical value makes rejection increasing in the numerator and decreasing in the independent denominator. This proves the comparison. It is a ceiling within this specified classical model and test class, not general optimality. An off-axis alternative has a noncentral denominator, so the ordinary noncentral- t power formula applies only to the aligned case.

19.5 Fixed simulation and all reference violations

Four fixed designs use $M \in \{12, 36\}$ periods, six units per period and serial correlation $\rho \in \{0, 0.6\}$. For entity index $i = 0, \dots, 5$, let z be a fixed standard Gaussian draw plus $0.25(i - 2.5)$, and let Φ contain six entity indicators, z and z^2 . The known covariance is the Kronecker product of the AR(1) correlation matrix and a six-unit equicorrelation matrix with correlation 0.4. For independent standard Gaussian innovations, generate

$$X = \Phi\gamma_X + L\epsilon_X, \quad Y = \Phi\gamma_Y + \tau L\epsilon_X + L\epsilon_Y, \quad X_0 = \Phi\gamma_X + L\epsilon_0. \quad (27)$$

Entity coefficients range linearly from -1 to 1 for X and from 0.5 to -0.5 for Y ; coefficients on (z, z^2) are $(2, 3)$ and $(0.3, 0.8)$ respectively. Effects are $\tau \in \{0, 0.15, 0.35\}$. The learned mean fits X_0 on the declared design. All methods have access to the same pilot scores and evaluation outcomes; no pilot outcome labels are supplied and no information-cost advantage is claimed.

Five scalar-adjustment directions use training-fold bin means at 8, 12, 16, 24, 32, both recorded extrapolation exponents, or the recorded cubic spline. These fit z only, so their uncalibrated normal/HAC results are adjustment probes, not replays of the main paper's entity-adjusted rank-panel procedure. Their Gaussian-reference versions all use the same full declared mean span and covariance, as do both leave-cluster directions and GLS. Thus calibrated direction comparisons have common model information.

Each fixed design has 1,000 independent innovation draws. Pairing three effects gives 12,000 distinct panels from 4,000 primitive draws. Each panel is also evaluated with the quadratic mean omitted and with dependence ignored. Those two reference violations are explicitly outside the theorem. All 468 method cells and 468,000 results are retained. This is one paired study, not 468 independent replications of generalization.

The 32 correctly specified finite-reference null rates range from 3.5% to 6.3%; all simultaneous Bonferroni 99% binomial intervals include 5%. This is an implementation diagnostic supporting the already specified theorem, not evidence that its premises hold elsewhere. The independent verification reconstructs every reported statistic and probability from the frozen primitives using separate matrix solves and reference formulas. The four primary paired learning comparisons use $\tau = 0.15$

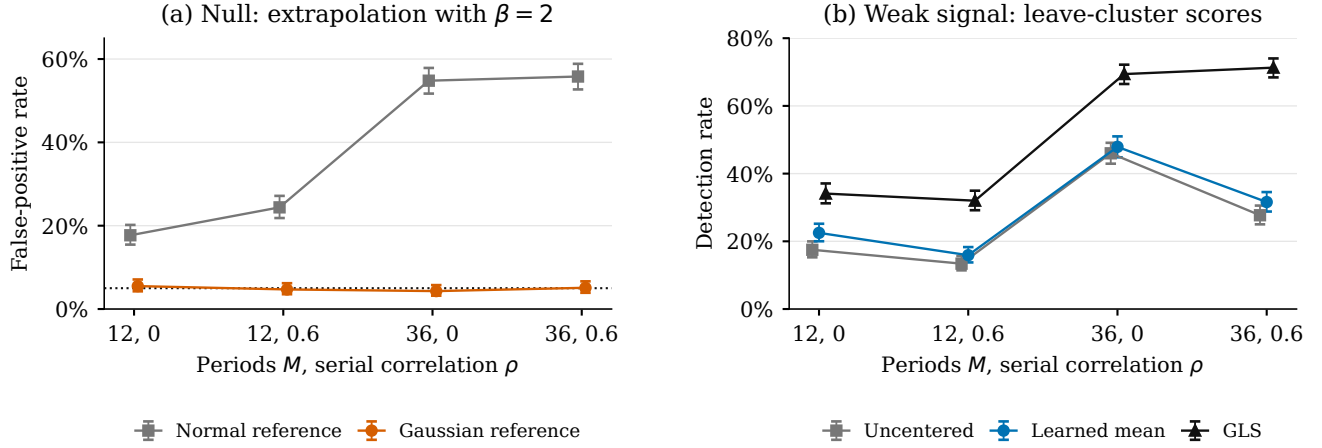


Figure 8: Four fixed designs under the correctly specified Gaussian model. Left: the scalar extrapolation direction with its original normal reference or the finite Gaussian reference. The dotted line is 5%. Right: weak-signal detection with uncentered or independently learned leave-cluster scores and GLS. Bars are marginal Wilson 95% intervals, not simultaneous intervals. Learning improves the chosen score but does not reach GLS. No real-panel calibration is inferred.

Table 15: Correct-reference weak-signal detection, $\tau = 0.15$; 1,000 paired draws per fixed design. Extrapolation uses $\beta = 2$. All entries use the same finite Gaussian reference. The full results also include nulls, stronger signals and deliberately incorrect references.

M	ρ	Extrapolation	Spline	Uncentered	Learned	GLS
12	0.0	.145	.138	.175	.225	.341
12	0.6	.110	.112	.134	.159	.320
36	0.0	.232	.292	.460	.479	.694
36	0.6	.164	.210	.277	.316	.713

and a one-sided exact binomial test of discordant decisions, with Holm correction across all four designs. Adjusted values are 0.000109, 0.00407, 0.01356 and 0.00371, respectively. They establish a gain over the uncentered leave-cluster direction in these models, not superiority to GLS. When dependence is ignored, finite-reference rejection rates across the outside-model null probes range from 5.1% to 19.7%. Omitting the quadratic mean term makes both learned leave-cluster and GLS directions reject in all 1,000 draws of each design. The restricted reference null is then false, despite conditional independence under the richer generator at $\tau = 0$. These results expose the consequences of asserting the wrong reference model.

20 Restricted-case results and their limits

These are retrospective audits of thirty stored prediction outputs in a UK insider-trading study. Each retained observation has the next calendar month’s return and a complete previous twelve-calendar-month return history, excluding the forecast month. The availability rule was fixed before recomputation, after examining the old results. It corrects source-data alignment; it is not a new prospective confirmation study.

The audit uses five contiguous whole-month folds. Adjustment is additive in firm and momentum-bin effects, at resolutions 8, 12, 16, 24, 32. The primary exponent is one, with two as sensitivity. Month clustering uses no between-month HAC and no copy or overlap guard. These settings differ from the guarded public screens. Normal-reference and BY calculations do not establish calibration.

Models differ in evaluation support, with 72 or 132 month clusters. Stored records, including repeated forecast records, are averaged within firm-month; source seed counts are one, four or five. Not every row has five predictions. Outcome encodings and stored precision differ; although subtracting a common monthly benchmark preserves ranks mathematically, float32 rounding can alter ties. This is not a common-sample or identical-target-array model ranking.

Historical feature preparation included firm-level forward fill followed by full-panel mean imputation. Removing incomplete audit records does not retrain forecasters, establish release times of all inputs, or verify foundation-model pretraining dates. These results therefore concern retrospective stored outputs, not deployable forecasting gains or implementable profits.

Both exponents retain zero models after BY. The adversarial residual-ranking MLP is the only nominal positive, with $T_1 = 2.453289$, $p_1 = 0.00707783$, and adjusted $p_1 = 0.848275$. For exponent two, $T_2 = 2.483673$, $p_2 = 0.00650175$, and adjusted $p_2 = 0.779232$. Its 31,045 observations span 1,852 firms and 132 months. Non-rejection establishes neither independence nor absence of predictive information. Historical full-support and outcome-only corrected aggregates are also supplied; each has one nominal positive and zero BY retentions.

Only model-level statistics, counts and methodological limitations are included. The underlying observations, entity identifiers, dates, predictions, residuals, monthly totals, training data and private loaders remain undistributed.

Table 16: All thirty restricted-case models on calendar-complete evaluation records. n is the number of firm-months; M counts months. Adjusted values use the entire thirty-model family, separately for each exponent. Every adjusted value exceeds 0.05.

Model	n	M	T_1	BY p_1	T_2	BY p_2
Adversarial residual-ranking MLP	31045	132	2.453	0.848	2.484	0.779
Chronos-Bolt-tiny	21428	132	0.561	1.000	0.591	1.000
Chronos-Bolt-small	21428	132	0.607	1.000	0.640	1.000
Chronos-Bolt-base	21428	132	0.452	1.000	0.485	1.000
Chronos-Bolt-base (extended coverage)	23881	132	-0.034	1.000	0.001	1.000
Moirai-1.0-R-base	21428	132	-3.542	1.000	-3.546	1.000
Moirai-1.0-R-large	21428	132	-3.372	1.000	-3.358	1.000
Untrained forecasting network	21428	132	-3.033	1.000	-2.988	1.000
WPMixer	8922	132	-2.324	1.000	-2.355	1.000
MambaSimple	8922	132	-2.219	1.000	-2.231	1.000
MSGNet	8922	132	-1.967	1.000	-1.991	1.000
DLinear	14770	132	0.107	1.000	0.101	1.000
FiLM	2613	132	-2.644	1.000	-2.671	1.000
TiDE	2613	132	-0.650	1.000	-0.619	1.000
SegRNN	2613	132	-0.625	1.000	-0.664	1.000
PatchTST	8113	72	-3.230	1.000	-3.227	1.000
iTransformer	8113	72	-4.002	1.000	-4.024	1.000
TimesNet	8113	72	-1.221	1.000	-1.232	1.000
TimeMixer	8113	72	-3.694	1.000	-3.682	1.000
TimeXer	8113	72	-3.990	1.000	-3.987	1.000
TFT	15699	132	-0.594	1.000	-0.577	1.000
NHITS	8690	132	-1.107	1.000	-1.134	1.000
DTAF	8113	72	-4.602	1.000	-4.582	1.000
DUET	16014	132	-2.185	1.000	-2.184	1.000
FreqCycle	8113	72	-3.787	1.000	-3.779	1.000
TimeBridge	16014	132	-0.955	1.000	-0.943	1.000
FT-Transformer	31045	132	-6.863	1.000	-6.863	1.000
TabM	31045	132	-3.315	1.000	-3.316	1.000
ImplicitDecoding	8113	72	-3.023	1.000	-3.017	1.000
TimeFilter	16014	132	-1.538	1.000	-1.541	1.000

21 A common-support temporal reconstruction comparison

The historical thirty-model table above is not a common-sample comparison and does not repair training-time information use. We therefore conduct a separate reconstruction experiment with the actual adversarial residual-ranking MLP, Ridge regression, histogram gradient boosting and an unfit momentum control. It retains 18,036 identical evaluation records and 132 common months. No month is excluded for a constant score or insufficient comparison support. This is a descriptive retrospective comparison, not a fresh prospective holdout.

Eleven expanding annual folds use the previously specified evaluation years. At each fit, training origins and next-calendar-month target maturity must both precede or equal the cutoff. Missing labels are excluded, never filled. All models use the same twelve available features: preceding one-, three-, six- and twelve-month returns, preceding twelve-month volatility, the transform $\log(1 + \text{count})$ for all, buy and sell disclosures, buy share, mean and maximum disclosure delay, and same-day disclosure share. The exact features are reconstructed independently from the restricted source records. Finite training medians fill features, with zero only for an entirely missing training column; means and population standard deviations are then fitted on training rows. Zero training scales become one.

The neural model uses its existing raw-rank and residual-rank squared losses, gradient reversal for issuer and month heads, and training-group stability penalty. The loss weights are 0.25, 1, 0.1 and 0.1 respectively; the additional worst-group and conditional-value-at-risk terms are disabled. Predictions combine the raw and residual heads with weights 0.25 and 1. The network has two hidden layers of width 128 with layer normalization and GELU activations, dropout 0.1, batch size 1,024, AdamW learning rate 0.001, weight decay 0.0001 and gradient clipping at five. All 20 epochs are run for seeds 11, 23 and 47 in each fold. The primary neural prediction is the fixed arithmetic mean of these seeds. There is no evaluation-based early stopping or parameter selection.

Ridge uses penalty ten. Histogram gradient boosting uses squared-error loss, learning rate 0.05, 200 iterations, at most fifteen leaves, at least thirty training samples per leaf, regularization one and 255 bins, without early stopping. Both fit the same within-training-month rank target as the raw neural head. The unfit control uses preceding one-month return. Fixed settings do not imply exhaustively optimized comparators.

All 55 fits finish: 33 neural, eleven Ridge and eleven boosting fits. The primary metric gives each common month equal weight and then averages its Spearman correlation. Pooled correlation answers a different question. Table 17 retains both metrics and every neural seed. The custom objective does not yield a descriptive advantage in this comparison. No significance test, confidence interval, model selection or finite certificate is inferred from these correlations.

Table 17: Temporal reconstruction on identical support. Correlations are descriptive. The three-seed mean is fixed before fitting; individual seeds are retained. All models share 18,036 records and 132 monthly evaluations.

Model	Mean monthly	Pooled	Monthly SD
Adversarial residual-ranking MLP, mean	.002746	-.000346	.090433
MLP, seed 11	.002577	.013670	.090463
MLP, seed 23	-.000632	-.014784	.102442
MLP, seed 47	.003744	-.002502	.090673
Ridge regression	.056668	.040327	.129204
Histogram gradient boosting	.037945	.022672	.117709
One-month momentum	.023142	-.004441	.122953

Independent checks rebuild every aggregate correlation, verify all training cutoffs and transformations, and reproduce all 33 neural and eleven Ridge checkpoints exactly. Boosting predictions and metrics are independently checked, but those models are not refitted in that verification. No raw observations, date rows, entity identifiers, forecasts, fitted transformation values, checkpoints or loaders are distributed; only model-level aggregates and protocol descriptions are supplied.

Historical availability remains unresolved. Transaction uniqueness is filtered across the current full extract; identifiers and adjusted prices also come from current source fields. Population selection is consequently retrospective despite verified training-only transformations and label maturity. The twelve-feature experiment neither repairs every historical model nor verifies vendor vintages, input release times, a new-security population or implementable profits.

Table 18: Feedback checks on two fixed observation patterns. Outcomes are simulated under the conditional null. Both statistics use entity and period clusters; the ordered version evaluates middle folds. Entries are the largest absolute mean and largest positive rejection rate across the stated cells, each with 200 replications. These two maxima need not occur in the same cell.

Design	Standard folds		Ordered folds	
	mean	Rate	mean	Rate
31,045 rows, six cells	3.092	0.890	0.251	0.070
8,113 rows, six cells	1.686	0.505	0.303	0.095
One-month lookback, one cell	2.237	0.735	0.242	0.090

22 New design checks and reproduction map

An executable independent-entity benchmark The script `equal_entity_directional.py` uses entirely synthetic trajectories, deterministic folds and equal entity weights to examine the raw-score corollary in the main paper. It includes complementary, directional and simple gapped-complementary means on the same middle rows. The gapped rule excludes a lookback’s width on either side of each evaluation fold; it is not a complete implementation of a previously published estimator. There are 500 paired replications in each of 84 cells. Entity counts are 25, 100 and 400; trajectory lengths 20 and 40; lookbacks one and five. Positive controls inject 0.1 or 0.3 times the current outcome innovation into the score and are not prospective forecasts. With 400 entities, 20 observations and lookback five, complementary, gapped and directional mean statistics are -2.677 , 3.148 and -0.002 , respectively; positive rejection counts are 0, 458 and 27 out of 500. Directional null rates across the complete grid range from 0.054 to 0.088, so finite-sample calibration is not exact. The package includes all cell summaries and replication-level synthetic statistics, exact expected means, standard-error comparisons and intervals.

The directory `results/design.validation/` contains the peer-sampling, feedback, interaction, near-copy, conditional-power, clustering, runtime, bootstrap and pretrained-rerun aggregates. Its README describes the design and unit of every table. The MDE values are measured increases in mean monthly rank correlation; the injected coefficient in the conditional-power experiment is a different quantity. Null quantiles fitted on the same simulated null replications do not constitute independent calibration.

The synthetic script `panel.design.checks.py` verifies the exact feedback identities for general weights and contiguous folds, the closed form for finite lookback, a serial-dependence counterexample, and fixed- versus resampled-peer expectations including ties. The directional audit script additionally reports complementary and directional fits on both all folds and matched middle folds. It retains original bin assignments and cohort-level copy flags; empty training sides and entities absent from a training side are reported explicitly. User-cluster ordering in the ratings task is not time ordering.

For continuous margins, the rank-reference term also has the classical representation $\Gamma = (3\tau - \rho_S)/12$. In a tie-free group of size N , let J average $a_{ij}b_{ij}$ over ordered distinct pairs and Q average $a_{ij}b_{i\ell}$ over ordered distinct triples. Then $J = (1 + \hat{\tau})/4$, $Q = (3 + \hat{\rho}_3)/12$, and

$$C_g - D_g = \frac{3\hat{\tau} - \hat{\rho}_3}{12(N-1)}, \quad \hat{\rho}_S = \frac{(N-2)\hat{\rho}_3 + 3\hat{\tau}}{N+1}.$$

Here $\hat{\rho}_3 = 12Q - 3$ is the degree-three U-statistic version of Spearman’s rho. This classical decomposition is attributed to the rank-statistic literature [4, 5]; it is not claimed as a new rank-correlation identity.

23 Paired factorial attribution of certificate efficiency

This section completes the missing absolute-allowance plus variance-bound comparison on the already inspected synthetic streams. It is a *post-exposure ablation*, not an independent new confirmation. The additional analysis protocol fixes all original settings and the reported paired contrasts before the added results are computed. The original study and every previously published result remain available.

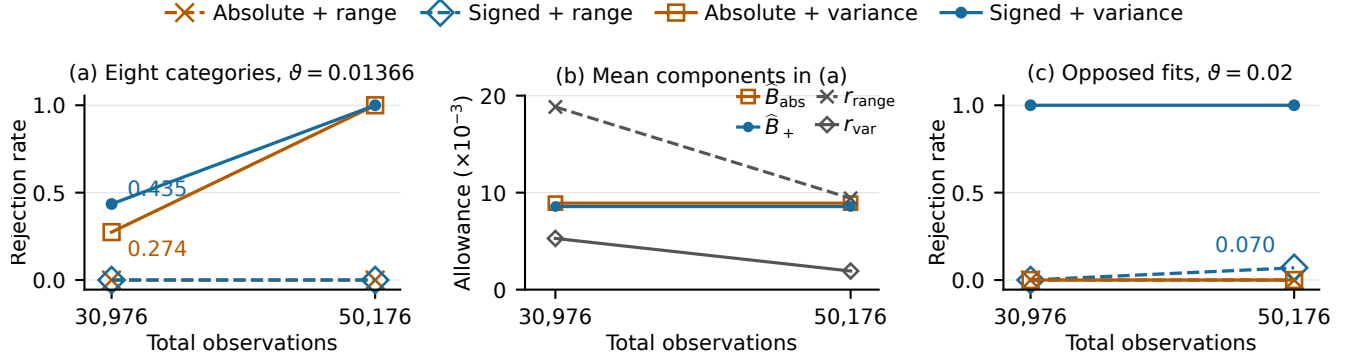


Figure 9: Complete factorial comparison. Four constructions share observations and 1,000 repetitions: (a) eight-category detection; (b) mean learning and sampling allowances; (c) detection with opposite fitted-mean shifts in two categories. Training uses 8,192 rows, validation 8,192 pairs, and 64 peers per group. Both pair members count toward cost; lines join only tested budgets.

Target and common information. Writing R_X, R_Y for population marginal midranks, including ties, the common target is

$$\vartheta = \mathbb{E}[(R_X - \mathbb{E}[R_X | C])(R_Y - \mathbb{E}[R_Y | C])].$$

Conditional on independent fitted category means f_c, g_c , the same order-three full-U center $\hat{U}$ has expectation $\vartheta + b_H$, where $b_H = \sum_c \pi_c (\mu_{X_c} - f_c)(\mu_{Y_c} - g_c)$. The four lower bounds are

$$L_{s,t} = \hat{U} - \hat{B}_s - r_t, \quad s \in \{\text{abs}, +\}, \quad t \in \{\text{range}, \text{var}\}.$$

They share all training, validation and evaluation draws, the fitted kernel range, and the full-U center. The two variance constructions also share the same sample variance of disjoint symmetric triples and the same $J = M \lfloor N/3 \rfloor$. Only the original absolute allowance is substituted for the signed allowance in the newly completed cell. We abbreviate the four constructions AR, SR, AV and SV in that order.

The absolute comparator is exactly the original one. On the common validation rectangles, put

$$e_c = \max_{x \in \{L_{X_c}, U_{X_c}\}} |x - f_c| \max_{y \in \{L_{Y_c}, U_{Y_c}\}} |y - g_c|.$$

Then $\hat{B}_{\text{abs}} = \max_{\pi \in \mathcal{P}} \sum_c \pi_c e_c$, where $\mathcal{P}$ is the original capped probability simplex. The signed allowance instead uses the largest signed corner product, with the identical rectangles, probability caps and allocation of δ . Empirical category weights, a worst-cell maximum and an absolute value of the summed product would be different comparators. The range and variance radii retain their original constants and error allocation; the variance inversion's factor two is retained. No assertion is made that these two sampling bounds are pointwise ordered for arbitrary inputs.

Complete grid and observation cost. All 100 original settings and 1,000 paired replications remain: the two- and eight-category laws, all five signals including zero and negative signals, $M \in \{25, 100, 400, 1600\}$, $N = 64$, both original validation budgets for estimated fits, and all original two-category opposed-shift settings. There are 400,000 method rows and 400 method-setting cells. The four constructions use 8,192 training rows, and their common charged cost is $8,192 + 2n_{\text{val}} + 64M$. The displayed budgets with $n_{\text{val}} = 8,192$ are 30,976 and 50,176. The original 40 nonpositive-target settings give 160 factorial cells; all have zero observed rejections. All 400 cells have zero observed lower-bound coverage failures. These observations do not imply zero risk.

Table 19: Completed paired factorial counts out of 1,000. All displayed cases use 8,192 validation pairs.

Law / fit	ϑ	Cost	AR	SR	AV	SV
Two / estimated	0.010	50,176	58	78	1,000	1,000
Eight / estimated	0.013655	30,976	0	0	274	435
Eight / estimated	0.013655	50,176	0	0	1,000	1,000
Two / opposed	0.020	30,976	0	0	0	1,000
Two / opposed	0.020	50,176	0	70	0	1,000

Paired intervals and attribution. For decisions A, B , let N_+ count $B = 1, A = 0$ and N_- count $B = 0, A = 1$. Each is marginally binomial, although the two counts are dependent. Two two-sided 97.5% Clopper–Pearson intervals for the discordant probabilities jointly cover with probability at least 95% by the union bound. Subtracting opposite endpoints therefore gives a conservative exact 95% interval for the paired probability difference. These are pointwise intervals for each fixed setting and contrast; there is no simultaneous claim for the 600 intervals in the artifact. They are neither independent-proportion intervals nor McNemar tests presented as confidence intervals.

Table 20: All six paired differences in the two highlighted estimated-fit settings. Entries are differences in rejection probability and conservative exact pointwise 95% intervals.

After minus before	Two, cost 50,176	Eight, cost 30,976
SR–AR	0.020 [0.007, 0.032]	0.000 [−0.004, 0.004]
AV–AR	0.942 [0.919, 0.957]	0.274 [0.238, 0.307]
SV–AR	0.942 [0.919, 0.957]	0.435 [0.395, 0.471]
AV–SR	0.922 [0.897, 0.940]	0.274 [0.238, 0.307]
SV–SR	0.922 [0.897, 0.940]	0.435 [0.395, 0.471]
SV–AV	0.000 [−0.004, 0.004]	0.161 [0.131, 0.189]

Thus the two-category increase from 58 to 1,000 is largely a sampling-bound improvement. The nonsaturated eight-category case isolates a further signed-allowance benefit under the variance bound: 161 favorable and zero unfavorable discordant replications, with 274 common rejections. Its paired increment is 0.161, with interval [0.131379, 0.188805]. At the lower opposed-fit budget, changing only the learning allowance under the range bound or only the sampling bound under the absolute allowance yields no additional decisions, whereas combining both yields 1,000; the SV–AV interval is [0.991255, 1].

Every continuous lower-bound difference decomposes exactly:

$$L_{+, \text{var}} - L_{\text{abs}, \text{range}} = (\hat{B}_{\text{abs}} - \hat{B}_+) + (r_{\text{range}} - r_{\text{var}}).$$

For the nonsaturated eight-category case the mean contributions are approximately 0.000354 from learning and 0.013557 from sampling. Mean absolute/signed allowances are 0.008918/0.008564 at both budgets; mean range/variance radii fall from 0.018838/0.005281 to 0.009419/0.001918. The training and validation draws are unchanged between these budgets. A small allowance difference can change decisions near the threshold, while thresholding itself creates interactions.

The paired interaction is $D = (SV - AV) - (SR - AR)$. Pointwise signed-allowance monotonicity implies $D \in \{-1, 0, 1\}$, so the same discordant-probability interval applies. For the two highlighted cases its values are respectively -0.020 with interval $[-0.032376, -0.007010]$, and 0.161 with interval $[0.131379, 0.188805]$. Order-averaged learning and sampling contributions have supports $[0, 1]$ and $[-1, 1]$. The artifact uses pointwise Hoeffding intervals with half-widths $\sqrt{\log(40)/(2n)}$ and $\sqrt{2\log(40)/n}$, respectively, and truncates to these known supports. For the eight-category case the contributions are 0.0805 [0.037553, 0.123447] and 0.3545 [0.268606, 0.440394]. Across the entire grid, the two factor orders give $1,348 + 10,217 = 8,340 + 3,225 = 11,565$ extra decisions. These totals are descriptive because different settings share draws; they have no pooled-binomial interval or unique causal allocation.

Persistent multicategory errors. For the actual two-category $\vartheta = 0.02$ opposed-shift setting, the true category means of both rank variables are (0.35, 0.65). In original training replication zero, the deployed shifted fits, rounded to eight decimal places, are

$$f \simeq (0.49982599, 0.80113030), \quad g \simeq (0.20166229, 0.49928231).$$

The unrounded fits give $b_H = -0.0225014272$ and $\mathbb{E}\hat{U} = -0.0025014272$. These are one random training replication, not fixed fits imposed on all replications. A concise algebraic illustration fixes $f = (0.5, 0.8)$ and $g = (0.2, 0.5)$, for which both

category product errors equal -0.0225 . At $\vartheta = 0.02$, vanishing uncertainty makes the signed lower bound tend to 0.02 but the absolute lower bound to -0.025 . In general the absolute limit loses $2 \sum_c \pi_c \{-(\mu_{Xc} - f_c)(\mu_{Yc} - g_c)\}_+$. This explains a persistent-error distinction; it does not establish superiority over fit-free conditional estimators.

Stronger classical comparisons and evidence scope. The artifact retains all 32 existing matched-comparator settings, 300 paired replications, eight methods and 128 specified contrasts. At the weaker two-category $\vartheta = 0.005$ setting with cost 30,976, SV, independent Bernstein, independent betting, pooled-U and pooled betting reject 1/10/263/2/273 out of 300; AV also rejects 1. Independent betting minus SV is 0.873333, with paired interval $[0.809435, 0.913159]$. The denominator differs from the 1,000-replication factorial table. Test-only betting has no plotted lower-confidence-bound radius. The classical Bernstein and betting constructions are not new methods, and these data refute a general dominance interpretation of the full-U variance certificate.

The portable artifact at `results/certificate_factorial` includes all 400,000 rows, the original 500,000-row inputs, all 48,000 matched-comparator input rows, exported scientific protocols and historical hashes, the pre-execution serialization-correction trace, and independent verification of all rows, all paired intervals, and selected original primitives. Its reproducible compact figure uses only settings declared before these added results. The publication rendering changes notation and leading zeros, without changing the data. The older public study and its broader grid remain available.

24 Public hourly forecast selection and chronological confirmation

This study compares association certificates with separately measured forecast losses on two public tasks. The forecasting rules, controls, inference methods, budgets and primary comparisons were fixed before source retrieval. The complete results include baseline fallbacks and unsuccessful primary comparisons. Independent sampling from each fixed selection archive supports its stated finite-archive association target; it does not make the original time series independent or certify a later forecast-loss target.

24.1 Sources, preparation and chronology

Appliances Energy Prediction, contributed by Luis Candanedo (2017), contains 19,735 ten-minute records [25]; its original ZIP is <https://archive.ics.uci.edu/static/public/374/appliances+energy+prediction.zip> (DOI <https://doi.org/10.24432/C5VC8G>). Metro Interstate Traffic Volume, contributed by John Hogue (2019), contains 48,204 records [26]; its ZIP is <https://archive.ics.uci.edu/static/public/492/metro+interstate+traffic+volume.zip> (DOI <https://doi.org/10.24432/C5X60B>). Both source datasets and the distributed derived hourly series, numeric forecast arrays and result tables are under CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>. Redistribution retains author attribution and records aggregation, reindexing, feature construction, clipping and model predictions as changes. Code is separately covered by the release’s MIT license. Original ZIP bytes and their SHA256 values are supplied; neither private data nor original model pickles are needed by this study’s public reproduction.

Appliances predicts energy use in the next complete nominal hour, in Wh, by summing six available ten-minute readings. An incomplete hour has a missing target. Metro predicts next-hour traffic volume at ATR301; duplicate nominal source timestamps are averaged. Its 7,629 duplicate source rows have no disagreements in the target. Reindexing the full recorded Metro calendar gives 52,551 hourly slots, of which 11,976 have no target. Missing slots remain missing; they are not removed before constructing lags. Source timestamps retain their nominal local interpretation without an unverified daylight-saving conversion.

For Appliances, T_0 is the first midnight at or after the first source timestamp, here 12 January 2016. The four half-open windows are $[T_0, T_0 + 56d)$, $[T_0 + 56d, T_0 + 70d)$, $[T_0 + 70d, T_0 + 84d)$ and $[T_0 + 84d, T_0 + 126d)$. For Metro, the nominal windows are before 1 January 2016, January–June 2016, July–December 2016 and calendar 2017; data from 2018 are unused. The source begins on 2 October 2012, so there are no artificial training rows before that first observation. Table 21 gives observed support. The fixed eligibility checks require at least 500 training targets and at least 75% of nominal hours in each later window; failure would be retained, without replacing the task or moving a window.

Table 21: Fixed chronological windows. Dates in the displayed ranges are inclusive; windows in the implementation are half-open at the next midnight.

Task	Stage	Observed target dates	Rows
Appliances	Training	12 Jan–7 Mar 2016	1,344
	Weight calibration	8–21 Mar 2016	336
	Selection	22 Mar–4 Apr 2016	336
	Confirmation	5 Apr–16 May 2016	1,008
Metro	Training	2 Oct 2012–31 Dec 2015	17,491
	Weight calibration	1 Jan–30 Jun 2016	3,476
	Selection	1 Jul–31 Dec 2016	4,362
	Confirmation	1 Jan–31 Dec 2017	8,713

Information time. Predictions for hour t use measurements from hours strictly before t , and known calendar variables. Completed measurements are *assumed* available at the end of their measurement hour, with no additional release delay. Historical data vintages and this publication-delay assumption have not been verified. Appliances’ interpolated airport variables and source random columns are excluded; Metro’s target-hour weather and weather text are excluded. Past confirmation outcomes can enter later lag features, as in an online forecast, while fitted models and selected rules stay fixed. The source parser necessarily materializes historical confirmation labels; the restriction concerns the fixed functions that produce forecasts and selections, not physical inability to access the downloaded file. Perturbing current and later targets leaves every feature through the current forecast origin unchanged in the implemented timing checks.

24.2 Forecasts, controls and frozen augmentation

All models use exact-calendar outcome lags 1, 2, 3, 6, 12, 24, 48, 168 hours; means, population standard deviations and observed-slot fractions over the preceding 24 and 168 slots (at least one observed slot); sine/cosine hour, weekday and day-of-year features; and a weekend indicator. The annual cycle uses 365.25 days. The richer feature set adds previous-hour Appliances temperatures T1–T9, humidities RH1–RH9 and lighting consumption, or Metro temperature, one-hour rain/snow and cloud coverage. Numeric covariates are aggregated at their own measurement hour and then shifted by one hour. Missing predictors receive training medians with missingness indicators; entirely missing training columns are retained with zero imputation.

The two baselines are the previous day’s same-hour target and Ridge on lag/rolling/calendar features. Ridge is the primary baseline. The eight candidates are persistence, daily seasonality, weekly seasonality, Ridge on the richer features, histogram gradient boosting, ExtraTrees, a category-only copy and independent uniform noise. Daily and weekly forecasts fall back to lag one and then the training median if needed. Ridge uses standardized imputed features, $\alpha = 10$ and an SVD solver. Boosting uses 200 iterations, 15 leaves, learning rate .05, minimum leaf size 20, $L_2 = .01$ and no early stopping. ExtraTrees uses 256 trees, maximum depth 12, minimum leaf size 10, all features and one execution thread. Boosting and ExtraTrees use the richer imputed feature matrix. The initial models are never refitted.

Let $s = \max\{1, \hat{Q}_{.99}(Y \mid \text{training})\}$ and $[x]_s = \min\{s, \max\{0, x\}\}$. Here $s = 2477.1$ Wh for Appliances and 6729.2 traffic units for Metro; the untruncated training medians used for fallback are 360 and 3349. Model fitting uses $[Y]_s$, and delivered forecasts are clipped to $[0, s]$. The primary loss is

$$\ell_s(y, p) = \{[y]_s - [p]_s\}^2 / s^2 \in [0, 1]. \quad (28)$$

Raw MSE and MAE compare the same delivered clipped prediction with the *untruncated* outcome; normalized MAE compares the clipped outcome and prediction, divided by s . Thus raw MSE is not merely the displayed bounded MSE multiplied by s^2 when outcome clipping is active.

For each baseline b and candidate f , calibration chooses the smallest minimizer of mean (28) for $[b + t(f - b)]_s$, with $t \in \{0, .05, \dots, 2\}$. A separate convex rule uses $t \in \{0, .05, \dots, 1\}$. All weights are frozen before selection metrics. The exact complete weight tables remain in the release. The selected ExtraTrees weights for Appliances/seasonal, Appliances/Ridge, Metro/seasonal and Metro/Ridge are respectively 1.10, 1.00, 1.05, 1.10; the four corresponding convex weights are all one.

Each baseline defines 32 declared categories: 16 training-baseline forecast quantile bins crossed with weekend status. All fifteen breakpoints are retained, including ties; equality is assigned to the lower bin. Breakpoints for fitted Ridge may use its training predictions, and are fixed before calibration/selection. Empty cells remain declared. Category-copy predictions are $s(c + .5)/32$ for category $c \in \{0, \dots, 31\}$, so they have exact zero conditional-rank target. These controls are stronger than four coarse calendar/sign cells but are still not the full continuous baseline or the full predictor feature vector.

The master seed is 2026091201. With task index $j = 0$ for Appliances and $j = 1$ for Metro, boosting and ExtraTrees use seeds $2026091201 + j + 10$ and $2026091201 + j + 20$; noise uses $2026091201 + j + 100$. Its draws are uniform on $[0, s]$ over the full recorded calendar. Sampling uses a NumPy `SeedSequence` with seed $2026091201 + j + 1000$, spawning three independent generators for training, validation and evaluation. The same task-level indices are reused across baselines, candidates and methods.

24.3 Sampling targets, five gates and complete decisions

Condition on the fitted forecasts and complete fixed selection archive of n rows, and let P_n be the uniform law on its rows. If $R_f(f) = P_n(f' < f) + \frac{1}{2}P_n(f' = f)$ is the marginal midrank, the four association gates target

$$\vartheta_n = E_{P_n}[(R_f - E_{P_n}[R_f \mid C])(R_y - E_{P_n}[R_y \mid C])]. \quad (29)$$

The rank inputs are unaugmented candidate forecasts and original outcome values; the selected augmentation is assessed on bounded loss separately. The direct-loss gate instead targets the fixed archive mean of $\ell_s(Y, b) - \ell_s(Y, [b + t(f - b)]_s)$, in $[-1, 1]$. Neither target is a later-period population-risk parameter.

Every gate uses 32,768 independently drawn uniform row indices with replacement. Reference U (RU) and reference betting (RB) allocate 4,096 to conditional-mean fitting, 8,192 independent pairs to validation and 12,288 to evaluation. RU uses the complete order-three distinct-reference center and the signed variance-sensitive bound; its kernel-variance sample uses 4,096 disjoint triples. RB uses the same validation allowance and independent triple scores with an equal mixture of 64 fixed geometric betting fractions from 10^{-4} to .99 [27]. Its reported value is $\min\{1, \delta + 1/E\}$, not an unadjusted e-BH procedure. A test-only betting result has no borrowed U-bound interval. For both methods the prespecified validation allocation is

$$\delta = \frac{0.1 \times .05}{8H_8}, \quad H_8 = \sum_{k=1}^8 k^{-1}. \quad (30)$$

Known-forecast U (KU) and known-forecast independent pairs (KP) additionally use the complete forecast and category metadata to center forecast midranks exactly, avoiding mean-learning error. Writing $R_X = F_X(X)$, $R_Y = F_Y(Y)$ and $m_X(Z) = \mathbb{E}[R_X | Z]$, exact centering gives

$$\mathbb{E}[(R_X - m_X(Z))R_Y] = \mathbb{E}[(R_X - m_X(Z))(R_Y - \mathbb{E}[R_Y | Z])] = \vartheta.$$

The omitted conditional-outcome mean contributes zero because $\mathbb{E}[R_X - m_X(Z) | Z] = 0$. This uses complete forecast-rank and category metadata; knowing only the baseline forecast does not supply it. Their classical Gini-covariance kernel [28] and fixed-frame range use all 32,768 sampled outcomes for evaluation: KU uses the complete order-two average and KP uses 16,384 disjoint pairs. Both retain ties. The direct-loss gate (LG) uses all 32,768 bounded gain observations and the classical independent-observation empirical-Bernstein bound [17]. KU, KP and LG have no validation-failure allocation. The release specifies all constants and inversions, including zero-range conventions.

Each of the five methods applies BY at .05 to the same complete eight candidates within each task/baseline; category-copy and noise are not removed after inspection. There is no pooled selection of the best gate and no claim of one global FDR guarantee over all methods and tasks. The rule chooses the retained augmentation with the smallest selection bounded MSE if it strictly improves on the baseline; otherwise baseline wins. Ungated augmentation (UG) compares all eight fitted augmentations without a gate; convex combination (CV) compares their separately calibrated convex versions. Baseline alone (B) completes the eight rules.

Table 22: All 32 fixed decisions and confirmation losses. ET denotes ExtraTrees augmentation and B the applicable baseline; t is its frozen weight ($t = 0$ on fallback). K_r is the number of BY-retained candidates for a gate, and a dash denotes a rule without a test. Norm. losses use the clipped training scale; raw losses use untruncated outcomes. All values are rounded only for display.

Rule	Forecast	t	K_r	Norm. MSE	Raw MSE	Norm. MAE	Raw MAE
<i>Appliances / seasonal baseline</i>							
RU	B	0.00	0	0.04129840	259,901.751	0.11114419	277.022
RB	B	0.00	0	0.04129840	259,901.751	0.11114419	277.022
KU	ET	1.10	5	0.01774890	114,034.668	0.08180081	204.335
KP	ET	1.10	6	0.01774890	114,034.668	0.08180081	204.335
LG	ET	1.10	6	0.01774890	114,034.668	0.08180081	204.335
UG	ET	1.10	–	0.01774890	114,034.668	0.08180081	204.335
CV	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
B	B	0.00	–	0.04129840	259,901.751	0.11114419	277.022
<i>Appliances / Ridge baseline</i>							
RU	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
RB	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
KU	ET	1.00	5	0.01727688	111,262.587	0.08055088	201.239
KP	ET	1.00	5	0.01727688	111,262.587	0.08055088	201.239
LG	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
UG	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
CV	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
B	B	0.00	–	0.02080396	132,473.153	0.10627496	264.960
<i>Metro / seasonal baseline</i>							
RU	B	0.00	0	0.02241721	1,016,372.888	0.08188022	552.156
RB	B	0.00	0	0.02241721	1,016,372.888	0.08188022	552.156
KU	ET	1.05	6	0.00143846	66,073.700	0.02440903	165.421
KP	ET	1.05	6	0.00143846	66,073.700	0.02440903	165.421
LG	ET	1.05	5	0.00143846	66,073.700	0.02440903	165.421
UG	ET	1.05	–	0.00143846	66,073.700	0.02440903	165.421
CV	ET	1.00	–	0.00140356	64,508.684	0.02396282	162.418
B	B	0.00	–	0.02241721	1,016,372.888	0.08188022	552.156
<i>Metro / Ridge baseline</i>							

Rule	Forecast	t	K_r	Norm. MSE	Raw MSE	Norm. MAE	Raw MAE
RU	B	0.00	0	0.00525213	238,913.748	0.05021910	339.102
RB	B	0.00	0	0.00525213	238,913.748	0.05021910	339.102
KU	ET	1.10	3	0.00137224	63,077.041	0.02370708	160.698
KP	ET	1.10	3	0.00137224	63,077.041	0.02370708	160.698
LG	ET	1.10	2	0.00137224	63,077.041	0.02370708	160.698
UG	ET	1.10	–	0.00137224	63,077.041	0.02370708	160.698
CV	ET	1.00	–	0.00140356	64,508.684	0.02396282	162.418
B	B	0.00	–	0.00525213	238,913.748	0.05021910	339.102

The reference methods fall back in all four task/baseline cases. KU and KP choose the same delivered ExtraTrees augmentation as UG in every case; LG also chooses it except for Appliances/Ridge, which falls back. For the primary Ridge comparisons, the augmentation reduces bounded MSE by 16.95% and 73.87%, and raw MSE by 16.01% and 73.60%, relative to baseline. Its *additional gain from gating is zero* because the ungated forecast is identical. Table 23 also gives the seasonal comparisons. CV differs because its calibration weights are restricted; with the seasonal baseline it slightly improves on UG in these confirmation windows. These are observed-support effects, not time-series confidence bounds.

Table 23: Confirmation MSE gains for the common KU/KP/UG forecast. Positive values favor that forecast. Its gain over UG is identically zero in all four rows. Gains against CV may have either sign.

Task	Baseline	Norm. gain vs B	Raw gain vs B	Norm. gain vs CV	Raw gain vs CV
Appliances	Seasonal	0.02354950	145,867.083	-0.00047202	-2,772.080
Appliances	Ridge	0.00352708	21,210.566	0.00000000	0.000
Metro	Seasonal	0.02097874	950,299.187	-0.00003490	-1,565.016
Metro	Ridge	0.00387988	175,836.707	0.00003132	1,431.644

The original primary criterion required RU to improve on *both* UG and CV under Ridge on *both* tasks. It is not met on either task (0/2); this criterion was not replaced by KU’s or KP’s better result. These are two new dataset tasks within the recorded exposure search, not proof that every researcher had never seen their data or that energy and transport had never appeared as broad subject domains. After the results were examined, new methods or configurations on these same windows would be exploratory rather than a new independent confirmation.

24.4 Full access cost and the learning-budget limitation

The raw source files, all observed targets, training forecasts, calibration losses and the complete selection archives are already materialized. The observed training/calibration/selection/confirmation supports total 3,024 Appliances and 34,042 Metro targets; the downloaded sources also contain unused rows. Source preparation builds forecasts over the recorded calendars, including unused periods. Model fitting and weight calibration are counted separately from rank-certificate queries. Of 32,768 replacement calls, the common stream touches all 336 Appliances selection rows and 4,361 of 4,362 Metro selection rows. The call count is respectively 97.52 and 7.51 times the selection census size. These repeated calls are not newly acquired labels. Streams are shared across methods/candidates, so multiplying calls by every method is not a physical acquisition cost either.

The known-forecast methods scan complete forecast/category metadata and allocate all sampled labels to evaluation; the reference methods have less exact forecast information and spend labels learning means and validating their errors. Equal label-call budgets therefore do not imply equal information or a one-component ablation. Computing the exact archive association uses the complete outcome census only as a diagnostic and is not supplied to a gate. Because selection already uses every selection label, no label-acquisition advantage over a census is established.

For ExtraTrees under the primary Ridge baselines, RU’s observed means are .00994791 and .00408298, its signed learning allowances .03810465 and .03704776, and its single-test .05 sampling radii .00519920 and .00505665. The exact finite-archive learning biases are only .00007855 and .00007659. All 32 unadjusted RU p-values and all 32 RB p-values are one, before BY. Holding these fitted means, validation samples and realized allowances fixed, adding evaluation observations alone cannot remove their allowance barrier. This does not prove failure of all redraws at the same budget. Substituting census bias after inspection is a diagnostic, not success of the fixed protocol or information available under its original training/validation allocation.

24.5 Four bounds allowing dependent confirmation hours

The four prespecified contrasts compare RU with UG and with CV, under Ridge on each task. Positive gain favors RU. These bounds concern a *same-period conditional availability-weighted average*; they are separate from (29) and from the descriptive MSE on observed rows. Let $T = 1008$ for Appliances and $T = 8760$ for Metro, including every nominal confirmation hour. Write I_t for target availability, and let $\mathcal{F}_{t-1}$ contain the training, calibration, selection and sampling information and all observations available before hour t . The selected prediction functions and their delivered predictions at t are $\mathcal{F}_{t-1}$ -measurable under the stated hour-end availability assumption. Set

$$D_t = I_t\{\ell_s(Y_t, p_t^{\text{comp}}) - \ell_s(Y_t, p_t^{\text{RU}})\} \in [-1, 1], \quad \mu_t = E[D_t \mid \mathcal{F}_{t-1}], \quad (31)$$

with $D_t = 0$ when the target is missing. An absent target is not imputed. The target is $\bar{\mu}_T = T^{-1} \sum_t \mu_t$, including availability inside the conditional expectation. No missing-at-random assumption, stationarity, independence of hourly observations or independence across the four contrasts is required.

For $\alpha_0 = .05/4 = .0125$, the reported lower bound is

$$L_T = \frac{1}{T} \sum_{t=1}^T D_t - \sqrt{\frac{2 \log(1/\alpha_0)}{T}}. \quad (32)$$

Conditional Hoeffding’s lemma [18], applied to a variable in an interval of width two, gives for $\lambda > 0$

$$E[\exp\{\lambda(D_t - \mu_t)\} \mid \mathcal{F}_{t-1}] \leq \exp(\lambda^2/2).$$

Iterating conditional expectations and using Markov’s inequality yields

$$P\left\{T^{-1} \sum_t (D_t - \mu_t) \geq r\right\} \leq \inf_{\lambda > 0} \exp\{-\lambda T r + T \lambda^2/2\} = \exp(-T r^2/2).$$

Substituting the radius in (32) gives $P\{L_T > \bar{\mu}_T\} \leq \alpha_0$. A union bound covers all four fixed conditional averages simultaneously with probability at least .95. This is a classical fixed-horizon bound, not an empirical-Bernstein time-series claim or an optional-stopping guarantee. The availability eligibility check does not license conditioning this coverage assertion on favorable observed support; failed tasks may not be silently removed.

Table 24: All four dependent confirmation bounds. Missing Metro hours contribute zero and the denominator is 8,760, not 8,713. Each bound has $\alpha_0 = .0125$; their joint coverage is at least .95 under the stated filtration. No lower bound is positive.

Task	Comparator	Nominal/observed hours	Calendar gain	Radius	Lower bound
Appliances	UG	1008/1008	-0.00352708	0.09324429	-0.09677137
Appliances	CV	1008/1008	-0.00352708	0.09324429	-0.09677137
Metro	UG	8760/8713	-0.00385907	0.03163009	-0.03548916
Metro	CV	8760/8713	-0.00382791	0.03163009	-0.03545801

The four nonpositive bounds give no positive conditional-average gain certificate; they are not a significance claim that the method harms forecasting. They do not cover an invariant future population, later years, unbounded raw squared loss or a rank-to-risk implication. The 944 weekly records in the release are descriptive summaries, not independent blocks to which an iid empirical-Bernstein formula is applied.

24.6 Exposure, mechanical correction and independent reproduction

The original local protocol and implementation were fixed at 17:02:15 UTC on 12 September 2026; source retrieval began afterward and completed at 17:02:29 UTC. This is a recorded local sequence, not an external preregistration or a trusted third-party timestamp. The forecasting/selection functions were fixed before the confirmation outcomes were inspected. Downloading and parsing all source labels does not create a claim that confirmation labels were physically inaccessible. All fixed tasks, methods and adverse outcomes are retained.

The first numerical implementation subtracted floating-point category means from forecast ranks. Category-constant scores acquired tiny nonzero residuals, which were amplified by a tiny computed kernel range, causing six erroneous category-copy rejections across KU and KP. The correction forms exact integer numerators: if $q_i = 2 \text{rank}(f_i) - 1$, category c has size n_c , and $S_c = \sum_{j: C_j=c} q_j$, then

$$A_i = \frac{n_{C_i} q_i - S_{C_i}}{2nn_{C_i}}.$$

Category-constant forecasts are now identically zero, including ties. The correction changes no model, category, split, seed, draw, coefficient or primary comparison. It removes the six false rejections; all 32 chosen rules and all four dependent confirmation bounds are unchanged. It was made after outcome exposure and is a mechanical replay of the same study, not a second independent confirmation. The defective computation is not on the public default reproduction path.

The public `results/forecast_confirmation` directory contains the original ZIPs, frozen numeric arrays, six complete result tables, protocol, source-to-export digest mapping, regression checks and separate producer and independent reconstruction code. Reproduction uses a new output directory and loads no historical model pickle. Python 3.12.11 with scikit-learn 1.7.2 rebuilds the original 70 forecast-array fields and eight sampling-index arrays exactly, and all non-runtime scientific table columns are exactly equal. The original fitting environment used scikit-learn 1.6.1; newly fitted models are saved for the installed version. Independent arithmetic additionally reconstructs all 160 gate rows, 32 choices, 32 confirmation losses, 32 candidate losses, 944 weekly rows and four dependent bounds without importing producer helpers. It checks source aggregation, scales, seasonal features, categories, weights, rank products, allowances, p-values and BY decisions. That independent arithmetic uses stated numerical tolerances, reported separately from the exact source-replay equality. No certificate is inferred from the mere agreement of two implementations.

25 Joint rank-mean constraints and an auditable numerical allowance

This section preserves the joint rank-mean allowance and records its complete exposed-archive diagnosis. The information added to the previous signed allowance is the exact mean of each population marginal midrank. Bilinear convexification, linear programming, weak duality and the sampling inequalities remain classical tools. The result is a conservative outer allowance, with no claim of sharpness or general optimality.

25.1 Original joint allowance and complete proof

The rectangular validation construction in the main paper gives intervals for each conditional midrank mean and category-mass caps u_c . Let q_c^+ be the largest endpoint product. This subsection retains the joint refinement and its proof, including its non-attainability example.

Lemma 2 (Joint rank-mean allowance). *Let (x_{ct}, y_{ct}) , $t = 1, \dots, 4$, be category c 's rectangle corners and $q_{ct} = (x_{ct} - f(c))(y_{ct} - g(c))$. Define*

$$\begin{aligned} B_{\text{joint}} &= \max_{\lambda \geq 0} \sum_{c,t} \lambda_{ct} q_{ct}, \\ \sum_t \lambda_{ct} &\leq u_c, \quad \sum_{c,t} \lambda_{ct} = 1, \\ \sum_{c,t} \lambda_{ct} x_{ct} &= \sum_{c,t} \lambda_{ct} y_{ct} = 1/2. \end{aligned} \tag{33}$$

On the same validation event, $b_H \leq B_{\text{joint}} \leq \hat{B}_+$.

Midranks average one half, including ties, coupling the category means. This classical bilinear relaxation [29] may replace $\hat{B}_+$ without another error budget. It is an outer bound, not generally attainable. Exact dual arithmetic verifies the numerical upper bound; empty sets or numerical failure return the separable bound.

Proof of Lemma 2. The identity $a(v, v') + a(v', v) = 1$ gives both marginal means $1/2$. On the validation event, bilinear interpolation expresses each true $(m_V(c), m_W(c), m_V(c), m_W(c))$ as a convex combination of the four lifted corners. Multiply its weights by p_c . The resulting λ is feasible in Eq. (33) and has objective b_H . Also $q_{ct} \leq q_c^+$, so every feasible objective is at most $\hat{B}_+$. Zero-mass categories and repeated corners cause no change. The set can be a strict relaxation: one category, intervals $[0, 1]$ and $f = g = 1/2$ permit $B_{\text{joint}} = 1/4$ although the true bias is zero. No extra validation event is used.

Population and confidence event Condition on independent training information. Write $p_c = \Pr(C = c)$, $\mu_{Xc} = E[R_X | C = c]$ and $\mu_{Yc} = E[R_Y | C = c]$, where R_X, R_Y are marginal population midranks under one common law. The comparison $a(x, x') = \mathbf{1}\{x > x'\} + \frac{1}{2}\mathbf{1}\{x = x'\}$ satisfies $a(x, x') + a(x', x) = 1$. Hence, including ties,

$$\sum_c p_c \mu_{Xc} = \sum_c p_c \mu_{Yc} = \frac{1}{2}. \tag{34}$$

References must be independent draws from the same global law. This identity does not assert that category-specific reference ranks have the same target. For fixed training fits $f_c, g_c \in [0, 1]$, the distinct-reference statistic has expectation $\vartheta + b$, with $b = \sum_c p_c (\mu_{Xc} - f_c)(\mu_{Yc} - g_c)$. Conditional independence of X and Y is not required.

Let the simultaneous validation event have probability at least $1 - \delta$, and on it let $\ell_c \leq p_c \leq h_c$, $L_{Xc} \leq \mu_{Xc} \leq U_{Xc}$ and $L_{Yc} \leq \mu_{Yc} \leq U_{Yc}$. All endpoints lie in $[0, 1]$. The archived comparison keeps the previous mean rectangles and one-sided mass caps, so $\ell_c = 0$; no validation observation or error allocation changes. The argument also permits valid mass lower bounds. Empty declared categories remain in the confidence set.

Compact corner-mass formulation Let (a_{cj}, b_{cj}) , $j = 1, \dots, 4$, be the four corners of the mean rectangle for category c , allowing repeated corners. Define

$$\begin{aligned} B_{\text{MC}} &= \max_{\lambda \geq 0} \sum_{c,j} \lambda_{cj} (a_{cj} - f_c)(b_{cj} - g_c), \\ \text{subject to } \sum_{c,j} \lambda_{cj} &= 1, \quad \ell_c \leq \sum_j \lambda_{cj} \leq h_c \quad \text{for all } c, \\ \sum_{c,j} \lambda_{cj} a_{cj} &= \sum_{c,j} \lambda_{cj} b_{cj} = \frac{1}{2}. \end{aligned} \tag{35}$$

The auxiliary corner masses are not additional observed categories. On the validation event, the product of the two endpoint interpolation weights, multiplied by p_c , represents the true pair of conditional means and their product. These masses are feasible and have objective b . If q_c is the largest corner error product, any feasible objective is at most $\sum_c q_c \sum_j \lambda_{cj}$. The category masses belong to the same probability confidence box as before. Therefore

$$b \leq B_{MC} \leq B_{sep}. \quad (36)$$

Repeated endpoints use a single interpolation endpoint; zero-mass categories use zero corner masses. Omitting the two half-mean equalities recovers B_{sep} exactly. Adding them spends no additional δ because Eq. (34) is a deterministic rank identity.

Equivalence to the perspective McCormick envelope Define three lifted mass moments, retaining h_c for the probability upper bound:

$$\xi_c = p_c \mu_{Xc}, \quad \zeta_c = p_c \mu_{Yc}, \quad \omega_c = p_c \mu_{Xc} \mu_{Yc}.$$

Besides the probability box and the global equalities, the relaxation imposes

$$\begin{aligned} L_{Xc} p_c &\leq \xi_c \leq U_{Xc} p_c, & L_{Yc} p_c &\leq \zeta_c \leq U_{Yc} p_c, \\ \omega_c &\geq L_{Yc} \xi_c + L_{Xc} \zeta_c - L_{Xc} L_{Yc} p_c, \\ \omega_c &\geq U_{Yc} \xi_c + U_{Xc} \zeta_c - U_{Xc} U_{Yc} p_c, \\ \omega_c &\leq L_{Yc} \xi_c + U_{Xc} \zeta_c - U_{Xc} L_{Yc} p_c, \\ \omega_c &\leq U_{Yc} \xi_c + L_{Xc} \zeta_c - L_{Xc} U_{Yc} p_c. \end{aligned} \quad (37)$$

The objective is $\sum_c (\omega_c - g_c \xi_c - f_c \zeta_c + f_c g_c p_c)$. For positive mass, division by p_c gives the classical bilinear hull on the rectangle [29]. More explicitly, for positive interval widths $d_X = U_X - L_X$ and $d_Y = U_Y - L_Y$, the high/high corner mass is

$$\lambda_{HH} = \frac{\omega - L_Y \xi - L_X \zeta + L_X L_Y p}{d_X d_Y}.$$

The high/low and low/high masses are respectively $(\xi - L_X p)/d_X - \lambda_{HH}$ and $(\zeta - L_Y p)/d_Y - \lambda_{HH}$; the remaining mass completes their sum to p . The four inequalities are exactly the nonnegativity of these four masses. If a width is zero, the corresponding product is fixed and ordinary one-dimensional interpolation suffices. If $p = 0$, all lifted coordinates vanish. Thus Eq. (35) and Eq. (37) are equivalent in these cases as well.

What the numerical program certifies A floating-point primal optimum is not used as an upper certificate. Write the lifted LP as $\max c^T x$ subject to $Ax \leq d_0$, $Ex = e_0$ and $0 \leq x \leq 1$. For any proposed multipliers $y \geq 0$ and arbitrary z , put $r = c - A^T y - E^T z$. Weak duality with the box residual gives

$$c^T x \leq d_0^T y + e_0^T z + \sum_i \max(r_i, 0) = U_{\text{dual}}. \quad (38)$$

The software reconstructs inputs, coefficients and proposed multipliers as exact binary rational numbers and evaluates this expression exactly. Approximate dual feasibility is not assumed. Negative residuals need no absolute-value correction. The returned allowance is the upward binary64 rounding of $\min(U_{\text{dual}}, B_{\text{sep}})$. It can exceed the mathematical optimum B_{MC} and is no larger than the similarly upward-rounded separable allowance. Exact fractions and multipliers are stored, so verification requires no optimizer.

Reported joint infeasibility or numerical failure returns the separable allowance. If the probability box itself is empty, the general API returns the universal bound $b \leq 1$. Genuine emptiness cannot occur on the stated confidence event and is not used as evidence for rejection. Malformed inputs raise an error. Exact LP arithmetic is conditional on valid input confidence intervals; it does not certify the floating-point evaluation of an upstream binomial quantile. In the archived run, the largest certified-upper correction relative to the floating primal objective was 1.87×10^{-14} .

If the existing sampling radius obeys $\Pr\{\bar{D} - (\vartheta + b) > r_\epsilon \mid \text{training, validation}\} \leq \epsilon$, the union bound gives $\Pr\{\bar{D} - B - r_{\alpha-\delta} > \vartheta\} \leq \alpha$ for the returned allowance B . No new sampling law, variance floor or conditional-independence assumption is introduced. The corresponding fixed-mixture betting construction keeps its existing additive δ correction; a validation-dependent capital is not thereby an unconditional e-value.

Strict improvement and strict relaxation An attainable strict improvement has two equally likely categories, $f = g = 0$, and identical rank-mean rectangles $[1/4, 3/8]$ and $[5/8, 3/4]$. Here $B_{\text{sep}} = 45/128$ and $B_{\text{MC}} = 5/16$. At all corners, $xy \leq (x + y)/2 - 3/16$, which with the global equalities proves the joint upper value. The law $X = Y = Z$ with Z uniform on $[0, 1]$ and categories indicating its two halves attains it.

Nevertheless, one category, both rectangles $[0, 1]$, and $f = g = 1/2$ permit half the corner mass at low/low and half at high/high. The LP gives $1/4$ although the two true conditional means are known to be $1/2$ and $b = 0$. General optimized corner weights need not be products of endpoint weights for a single coherent pair of conditional means.

A nontrivial example fixes category masses $(1/4, 3/8, 3/8)$, fits $f = g = 1/2$ and all mean rectangles $[3/8, 5/8]$. The relaxed optimum is $1/64$, whereas the maximum over coherent means is $3/256$. Writing each centered mean as $t_c/8$ with $|t_c| \leq 1$ and $p^T t = 0$, Cauchy–Schwarz reduces the latter maximum to a weighted squared norm. Enumerating the vertices of that sliced box gives maximum $3/4$, attained by $t = (0, 1, -1)$. It is attainable by a real rank law: let $X = Y = Z$ be uniform and set

$$\Pr(C = 1 \mid Z) = \frac{1}{4}, \quad \Pr(C = 2 \mid Z) = \frac{3}{8} + \frac{9}{16}(Z - \frac{1}{2}), \quad \Pr(C = 3 \mid Z) = \frac{3}{8} - \frac{9}{16}(Z - \frac{1}{2}).$$

These nonnegative propensities give means $(1/2, 5/8, 3/8)$. The relaxation gap is $1/256$. Further common-rank constraints, such as $p_c/2 \leq \mu_{Xc} \leq 1 - p_c/2$, are not imposed. These facts preclude calling the LP a sharp attainable rank-law optimization.

Complete post-exposure archive diagnosis The construction and checks were frozen before its first new allowance outputs, but after exposure to the forecast study. This is explicitly an exploratory diagnosis, not a new confirmation. It retains every one of the $2 \times 2 \times 8 = 32$ task/baseline/candidate cases, their original fits, 32 declared categories, probability caps, confidence allocation and draws. Per candidate, 4,096 training draws, 8,192 disjoint validation pairs and 12,288 evaluation draws cost 32,768 row queries. No census rank or conditional-mean metadata enters the new allowance. Those values are used only afterward for diagnostics.

All 32 allowances decrease: the median absolute reduction is 0.000479251, the maximum is 0.002510925, and their relative reductions have median 1.2399% and maximum 6.5173%. Every variance and betting raw p -value remains exactly one; all BY values remain one. Both reference methods therefore retain no candidate in any family. The four category-copy controls remain nonrejected. This does not recover the earlier failed practical success criterion or demonstrate a forecast-loss gain.

Table 25: ExtraTrees allowances in the complete exploratory diagnosis. The lower bounds remain negative and both reference decisions are unchanged.

Task	Baseline	Separable	Joint	Reduction (%)	Joint lower
Appliances	Daily seasonal	.0418754	.0413886	1.1626	−.0106669
Appliances	Ridge	.0381046	.0380884	.0427	−.0333397
Metro	Daily seasonal	.0392165	.0386157	1.5320	−.0287812
Metro	Ridge	.0370478	.0361197	2.5050	−.0370934

The public directory `results/joint.bias/` contains all 32 rows, their exact dual receipts and local numeric inputs. Running `python results/joint.bias/run.py` checks the bundle, reconstructs all allowances and probabilities, and invokes a separately implemented verifier that imports neither the producer nor an optimizer. It checks 200 additional finite rank laws and 240 exact corner-formulation conversions, including ties, zero masses and degenerate rectangles. Separate producing code checks cover 120 finite frames and numerical-failure branches. The reconstruction retains the previously verified full-U means and radii; it rebuilds the training and validation quantities, disjoint-triple kernels, variances, ranges and probability inversions. It does not repeat forecast training or confirmation outcomes. Numeric equality and tolerance-based agreement are reported separately, and all old sealed results are preserved.

26 Aggregate validation with exact category probabilities

Lemma 1 in the main paper bounds the total learning bias using independent validation streams. The category probabilities are known exactly, for example from the complete control column of a finite archive. Forecast and outcome ranks are not required to obtain these probabilities. Conditioning fixes only focal validation categories; reference observations and their category labels retain their common marginal law. Estimated category frequencies alone do not meet this information condition.

Validation experiment The study crosses $C \in \{2, 8, 32, 128\}$ categories, $m \in \{512, 2048, 8192\}$ validation pairs, two category distributions and three fitting-error patterns. It has 72 settings with 2,000 repetitions each, using $\delta = 0.05$ for each method separately. Balanced masses are $p_c = 1/C$; unequal masses are $p_1 = 0.7$ and $p_c = 0.3/(C - 1)$ for $c > 1$. Let t_c be equally spaced on $[-1, 1]$, recentered to have p -weighted mean zero, and set

$$q_c = \frac{1}{2} + 0.35 t_c / \max_d |t_d|.$$

Within a category, V, W are independent Bernoulli variables with success probability q_c . Reference pairs follow the common mixture law; their two channels therefore share reference-category dependence. Since $\sum_c p_c q_c = 1/2$, both conditional marginal-midrank means are $\mu_c = 1/4 + q_c/2$. The three fixed fits are $(f_c, g_c) = (\mu_c, \mu_c)$, $(\mu_c + 0.05, \mu_c + 0.05)$, and $(\mu_c + 0.15, \mu_c - 0.15)$, with true biases 0, 0.0025 and -0.0225 . All fits lie in $[0, 1]$; these patterns probe validation accuracy rather than a particular learning algorithm.

Both methods receive the same exact p and the same m independent focal/reference pairs, costing $2m$ raw observations. The aggregate method uses forecast comparisons from the first half and outcome comparisons from the second half. The rectangle comparator uses both channels from all pairs. If $n_c > 0$, its mean radius is $\sqrt{\log(4C/\delta)/(2n_c)}$, with intervals intersected with $[0, 1]$; missing categories use $[0, 1]$. It maximizes each error product over the four endpoints, then averages those maxima with p . Thus it receives both channels on more validation units and the same category metadata. Neither bound is selected by taking an unadjusted random minimum.

The primary generator uses the exact 16-outcome binary-pair law and multinomial counts to reproduce the independent-pair sampling distribution. A separate verifier checks 288 selected replications. A second sealed seed draws the raw focal and reference categories and their four Bernoulli variables directly: 1,600 repetitions yield 4,800 fit rows and 48 method summaries. The original fixed seed is 1309202601; the second is 1309202699. These checks concern coverage and width, not forecast selection power or a new chronological confirmation.

At $C = 32$, $m = 8192$ and balanced exact fits, the median slack above true bias is 0.0185168 for rectangles and 0.00260164 for aggregation. Across all settings, the ratio of rectangle to aggregate median slack has minimum 0.5574, median 2.5000 and maximum 16.9403. Rectangles have smaller median slack in 10/72 settings: all six $C = 2$ opposite-error settings, three $C = 2$ unequal-mass exact-fit settings, and the $C = 128$, $m = 512$, unequal-mass opposite-error setting. Every one of the 144 method cells has zero observed noncoverage among 2,000 repetitions; the second-seed cells also have zero observed noncoverage. These counts do not imply zero failure probability. Negative valid allowances are retained; they are not truncated to zero.

Matched archive diagnosis After the original Appliances and Metro task outcomes were inspected, the aggregate allowance was applied to all 32 existing task/baseline/candidate cases. Fits, forecasts, draws and evaluation scores remain fixed. Each candidate uses 4096 training observations, 8192 validation pairs and 12288 evaluation observations, totaling 32768 queries. Exact category masses come from the selection archive’s control column. The original rectangle additionally estimates masses; the matched rectangle and aggregate bounds receive identical exact masses. Each construction uses the original per-candidate validation allocation and is assessed separately. BY applies within each of the four eight-candidate families.

Table 26: All 32 inspected candidates. Counts combine four separate families; they do not count independent confirmations. The exact-bias row is diagnostic.

Learning allowance	Pooled U retained	Betting retained
Original rectangles, estimated masses	0	0
Rectangles, exact masses	1	3
Aggregate, exact masses	9	13
Exact archive bias	14	16

For Appliances ExtraTrees beyond Ridge, the matched rectangle allowance is 0.0292745 and the aggregate allowance is 0.00432772. The observed association is 0.00994791 and the pooled-U sampling radius is 0.00519920. Its pooled-U lower

bound is positive at the pointwise 0.05 level, but its BY value is 0.367573, so this rule does not retain it. Classical betting with the same aggregate allowance has BY value 0.00306193 and retains it. For Metro ExtraTrees beyond Ridge, the aggregate allowance 0.00573561 exceeds observed association 0.00408298; neither rule retains it. Substituting exact archive bias makes the Metro betting rule retain it, whereas the pooled-U rule still does not. Both validation width and sampling precision therefore matter. All category-copy and independent-noise controls remain unretained. The original prespecified two-task success result remains 0/2; the post-exposure diagnostic does not establish a new gating benefit.

Reproduction and scope The public directory `results/aggregate_bias/` contains all 128 diagnostic rows, 144,000 simulation rows, 144 summaries, both seeds, source-specific provenance and separately implemented checks. Its reproduction command reconstructs all archive rows and verifies simulation arithmetic; `--regenerate-simulation` also generates every original replication. Outputs are written outside the distributed evidence. Numerical comparisons use the stated floating-point tolerances; the theorem is a real-arithmetic statement, not an outward-rounded machine certificate.

Direct product-bias estimation has classical higher-order antecedents [30,31]. Active U-statistic inference optimizes label acquisition while incorporating predictions [32]; here category probabilities are given and validation queries are fixed. The present contribution is the explicit observable allowance for this rank target and information condition. It does not claim optimal acquisition, minimax width, sharp rank-law optimization or uniform improvement over all classical procedures.

Reference design values The reference comparison in the main figure uses resampled Gaussian peers with Kendall correlation 1/3 and Spearman correlation 0.482584. Its fixed-peer counterparts have shared-score mean zero and distinct-score means -0.001423 or $+0.001423$, according to channel-effect alignment. These design-specific values are separate from the aggregate validation study.

A sufficient training rate For fixed finite categorical controls with positive cell probabilities, empirical midranks followed by cell averaging in an independent training sample of size m give $\varepsilon_f, \varepsilon_g = O_p(m^{-1/2})$. Thus $m/\sqrt{MN} \rightarrow \infty$ suffices, counting all training observations. This states a usable sufficient rate; it is not a necessary cost bound.

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